

Acceleration and Deceleration Time [Equation (22-3)]

$$t = \frac{Wk_e^2(\Delta n)}{308T} = \frac{(11.83)(550)}{308(240)} = 0.088 \text{ s}$$

Cycling Rate and Heat Dissipation [Equation (22-6)]

For a total cycle time of 2.896 s, the number of cycles per minute would be

$$C = \frac{1.0 \text{ cycle}}{2.896 \text{ s}} \frac{60 \text{ s}}{\text{min}} = 20.7 \text{ cycles/min}$$

The energy generated with each engagement of either the clutch or the brake is

$$E = 1.7 \times 10^{-4} Wk_e^2 n^2 = 1.7 \times 10^{-4} (11.83)(550)^2 = 608 \text{ lb} \cdot \text{ft}$$

The energy generation per minute is

$$E_t = 2 EC = (2)(608 \text{ lb} \cdot \text{ft/cycle})(20.7 \text{ cycles/min}) = 25\,200 \text{ lb} \cdot \text{ft/min}$$

This is greater than the heat-dissipation capacity of unit G at rest (18 000 lb · ft/min). Then let's compute a weighted average capacity for this cycle. First, referring to Figure 22-15, approximately 1.735 s is "at speed," or 550 rpm. The balance of the cycle, 1.161 s, is at rest. From Table 22-1, and interpolating between zero speed and 1800 rpm, the heat-dissipation rate at 550 rpm is approximately 28 400 lb · ft/min. Then the weighted average capacity for unit G is

$$E_{\text{avg}} = \frac{t_0}{t_t} E_0 + \frac{t_{550}}{t_t} E_{550}$$

where t_t = total cycle time

t_0 = time at rest (0 rpm)

t_{550} = time at 550 rpm

E_0 = heat-dissipation capacity at rest

E_{550} = heat-dissipation capacity at 550 rpm

Then

$$E_{\text{avg}} = \frac{1.161}{2.896} (18\,000) + \frac{1.735}{2.896} (28\,400) = 24\,230 \text{ lb} \cdot \text{ft/min}$$

This is a little lower than required, and the design would be marginal. Fewer cycles per minute should be specified.

22-10 FRICTION MATERIALS AND COEFFICIENT OF FRICTION

Many of the types of clutches and brakes discussed in this chapter employ mating surfaces driven through friction materials. The function of these materials is to develop a substantial friction force when a normal force is created by the brake actuation means. The friction force produces a force or a torque that retards existing motion if applied as a brake or that accelerates a member at rest or moving at a slower speed if applied as a clutch.

The desirable properties of friction materials are as follows:

1. They should have a relatively high coefficient of friction when operating against the mating materials in the system. The highest coefficient of friction is not always the best choice, because smooth engagement is often aided by a more moderate frictional force or torque.

2. The coefficient of friction should be relatively constant over the range of operating pressures and temperatures so that reliable, predictable performance can be expected.
3. The materials should have good resistance to wear.
4. The materials must be chemically compatible with mating components.
5. Environmental hazards must be minimized.

Several materials are used for friction elements in clutches and brakes, and many are proprietary to a particular manufacturer. Common in the past were various asbestos-based compounds having coefficients of friction in the range of 0.35 to 0.50. Asbestos has been shown to be a health hazard and has been replaced by molded compounds of polymers and rubber. Where flexibility is required, as in band brakes, the base material is woven into a fabric, sometimes strengthened with metal wire,

TABLE 22-2 Coefficients of Friction

Friction material	Dynamic friction coefficient		Pressure range	
	Dry	In oil	(psi)	(kPa)
Molded compounds	0.25–0.45	0.06–0.10	150–300	1035–2070
Woven materials	0.25–0.45	0.08–0.10	50–100	345–690
Sintered metal	0.15–0.45	0.05–0.08	150–300	1035–2070
Cork	0.30–0.50	0.15–0.25	8–15	55–100
Wood	0.20–0.45	0.12–0.16	50–90	345–620
Cast iron	0.15–0.25	0.03–0.06	100–250	690–1725
Paper-based		0.10–0.15		
Graphite/resin		0.10–0.14		

saturated with a resin, and cured. Cork and wood are also used sometimes. Paper-based materials are used in some oil-filled clutches. In severe environments, cast iron, sintered iron or other metals, or graphite materials are employed. Table 22-2 gives approximate ranges of coefficients of friction and the pressure that the materials can withstand.

For automotive applications, standards are set by SAE International. In SAE J866 (Reference 7), a set of codes is defined to classify friction materials according to the coefficient of friction, regardless of the material used. Table 22-3 lists these codes.

In problems in this book that require a coefficient of friction, we will use a value of 0.25 unless otherwise stated. Based on the values reported here, this is a relatively low value that should yield conservative designs.

Materials for Discs and Drums

A variety of metals are applied in the manufacture of brake and clutch discs and drums. The material must have sufficient strength, ductility, and stiffness to withstand the applied forces while maintaining a precise

geometry. It must also be able to absorb heat from the friction surface and dissipate it to the environment.

Some of the popular choices are gray cast iron, ductile iron, carbon steel, and copper alloys. Many discs and drums are cast for cost reasons and to obtain near net shape of the parts that require little machining after casting. Cast iron has low cost and high thermal conductivity as compared with ductile iron. However, ductile iron is better able to withstand shock or impact loading. Copper alloys have far higher thermal conductivity than the other materials, but they have poorer wear performance.

22-11 PLATE-TYPE CLUTCH OR BRAKE

Figure 22-2(a) shows a simple sketch of a plate-type clutch, and Figure 22-8 shows a cutaway view of a commercially available, electrically actuated clutch or brake. In Figure 22-8, an electromagnet exerts an axial force that draws the friction surfaces together. When two bodies are brought into contact with a normal force between them, a friction force that tends to resist relative motion is created. This is the principle on which the plate-type clutch or brake is based.

Friction Torque

As the friction plate rotates in relation to the mating plate with an axial force pressing them together, the friction force acts in a tangential direction, producing the brake or clutch torque. At any point, the local pressure times the differential area at the point is the *normal force*, N . The normal force times the coefficient of friction is the *friction force*. The friction force times the radius of the point is the *torque* produced at this point. The *total torque* is the sum of all of the torques over the entire area of the plate. We can find the sum by integrating over the area.

There is usually some variation of pressure over the surface of the friction plate, and some assumption about

TABLE 22-3 Coefficient of Friction Classification Codes of the Society of Automotive Engineers

Code letter	Coefficient of friction
C	Not over 0.15
D	Over 0.15 but not over 0.25
E	Over 0.25 but not over 0.35
F	Over 0.35 but not over 0.45
G	Over 0.45 but not over 0.55
H	Over 0.55
Z	Unclassified

the nature of the variation must be made before the total torque can be computed. A conservative assumption that yields a useful result is that the friction surface will wear uniformly over the entire area as the brake or clutch operates. This assumption implies that the product of the local pressure, p , times the linear relative speed, v , between the plates is constant. Wear has been found to be approximately proportional to the product of p times v .

Taking all these factors into account and completing the analysis produces the following result for friction torque:

$$T_f = fN(R_o + R_i)/2$$

But the last part of this relation is the mean radius, R_m , of the annular plate. Then

Friction Torque on Annular Plate

$$T_f = fNR_m \quad (22-8)$$

As stated before, this is a conservative result, meaning that the actual torque produced would be slightly larger than predicted.

Wear Rating

Note that the torque is proportional to the mean radius, but that no area relationship is involved in Equation (22-8). Thus, the completion of the design for final dimensions requires some other parameter. The missing factor in Equation (22-8) is the expected wear rate of the friction material. It should be obvious that even with the same mean radius, a brake with a larger area would wear less than one with a smaller area.

Manufacturers of friction materials can assist in the final determination of the relationship between wear and the area of the friction surface. However, the following guidelines allow the estimation of the physical size of brakes and will be used for problem solutions in this book.

The wear rating, WR , will be based on the frictional power, P_f , absorbed by the brake per unit area, A , where

Frictional Power

$$P_f = T_f \omega \quad (22-9)$$

and ω is the angular velocity of the disc. In SI units, with torque in $\text{N} \cdot \text{m}$ and ω in rad/s , the frictional power is in $\text{N} \cdot \text{m/s}$ or watts. In the U.S. Customary System, with torque in $\text{lb} \cdot \text{in}$ and angular velocity expressed as n rpm, the frictional power is in hp, computed from

$$P_f = \frac{T_f n}{63\,000} \text{ hp} \quad (22-10)$$

For industrial applications, we will use

Wear Rating

$$WR = P_f/A \quad (22-11)$$

where $WR = 0.04 \text{ hp/in}^2$ for frequent applications, a conservative rating

$WR = 0.10 \text{ hp/in}^2$ for average service

$WR = 0.40 \text{ hp/in}^2$ for infrequently used brakes allowed to cool somewhat between applications

Example Problem 22-6

Compute the dimensions of an annular plate-type brake to produce a braking torque of $300 \text{ lb} \cdot \text{in}$. Springs will provide a normal force of 320 lb between the friction surfaces. The coefficient of friction is 0.25 . The brake will be used in average industrial service, stopping a load from 750 rpm .

Solution

Step 1. Compute the required mean radius. From Equation (22-8),

$$R_m = \frac{T_f}{fN} = \frac{300 \text{ lb} \cdot \text{in}}{(0.25)(320 \text{ lb})} = 3.75 \text{ in}$$

Step 2. Specify a desired ratio of R_o/R_i and solve for the dimensions. A reasonable value for the ratio is approximately 1.50 . The range can be from 1.2 to about 2.5 , at the designer's choice. Using 1.50 , $R_o = 1.50R_i$ and

$$R_m = (R_o + R_i)/2 = (1.5R_i + R_i)/2 = 1.25R_i$$

Then

$$R_i = R_m/1.25 = (3.75 \text{ in})/1.25 = 3.00 \text{ in}$$

$$R_o = 1.50R_i = 1.50(3.00 \text{ in}) = 4.50 \text{ in}$$

Step 3. Compute the area of the friction surface:

$$A = \pi(R_o^2 - R_i^2) = \pi[(4.50 \text{ in})^2 - (3.00 \text{ in})^2] = 35.3 \text{ in}^2$$

Step 4. Compute the frictional power absorbed:

$$P_f = \frac{T_f \dot{\theta}}{63\,000} = \frac{(300 \text{ lb} \cdot \text{in})(750 \text{ rpm})}{63\,000} = 3.57 \text{ hp}$$

Step 5. Compute the wear rating:

$$WR = \frac{P_f}{A} = \frac{3.57 \text{ hp}}{35.3 \text{ in}^2} = 0.101 \text{ hp/in}^2$$

Step 6. Judge the suitability of WR . If WR is too high, return to Step 2 and increase the ratio. If WR is too low, decrease the ratio. In this example, WR is acceptable for average service.

A more compact unit can be designed if more than one friction plate is used. We multiply the friction torque for one plate by the number of plates to determine the total friction torque. One disadvantage of this approach is that the heat dissipation is relatively poorer than for the single plate.

Improving Wear Performance of Brakes

Actual wear in a given application depends on a combination of many variables. Friction materials are relatively softer and weaker than the metallic materials used for discs and drums. Wear is often characterized as adhesion. As the surface of the friction material rubs over the high spots of the metal, plastic deformation occurs at the surface and particles are sheared off, breaking the bond between particles or dislodging filler materials from the polymer bonding agents. This process accelerates when surface temperatures rise as the brake absorbs the energy required to stop the rotating system. The thermal behavior of the system is critical to good life. If temperatures rise above 400°F (200°C), wear rate increases significantly and the coefficient of friction decreases, leading to poorer braking performance called *fade*.

It is difficult to predict the life of a given brake system analytically, and testing under real operating conditions is recommended for new designs. The following lists the general principles for improving wear performance:

- Specify friction materials that have relatively low adhesion when in contact with the disc or drum material.
- Specify friction materials that have high bonding strength between constituent particles.
- Provide high hardness on the surface of the disc or drum by heat treatment.
- Keep the pressure between the friction material and the material of the disc or drum as low as practical.
- Maintain the surface temperature at the interface between the friction material and the material of the disc or drum as low as practical by promoting heat transfer away from the system by conduction, convection, and radiation. Forced airflow or cooling with water is often applied in critical situations.
- Provide a smooth surface finish on the discs and drums.

- Provide lubricants such as oil or graphite at the friction interface.
- Exclude abrasive contaminants from the friction interface.
- Minimize slipping between the clutch or brake elements by promoting lockup of the engaging elements.

22-12 CALIPER DISC BRAKES

The disc brake pads are brought into contact with the rotating disc by fluid pressure acting on a piston in the caliper. The pads are either round or in a short crescent shape to cover more of the disc surface [see Figures 22-2(b) and 22-7]. However, one advantage of the disc brake is that the disc is exposed to the atmosphere, promoting the dissipation of heat. Because the disc rotates with the machine to be controlled, heat dissipation is further enhanced. The cooling effect improves the fade resistance of this type of brake relative to the shoe-type brake.

Designs for the friction torque and for the wear rating are similar to those explained for plate-type brakes.

22-13 CONE CLUTCH OR BRAKE

The angle of inclination of the conical surface of the cone clutch or brake is typically 12°. A lower angle could be used with care, but there is a tendency for the friction surfaces to engage suddenly with a jerk. As the angle increases, the amount of axial force required to produce a given friction torque increases. Thus, 12° is a reasonable compromise.

Referring to Figure 22-16, we see that as an axial force F_a is applied by a spring, manually or by fluid pressure, a normal force N is created between the mating friction surfaces, all around the periphery of the cone. The desired friction force F_f is produced in the tangential direction, where $F_f = fN$. It is assumed that the friction force acts at the mean radius of the cone so that the friction torque is

$$T_f = F_f R_m = fNR_m \quad (22-8)$$

In addition to the tangentially directed friction force, a friction force develops along the surface of the cone and

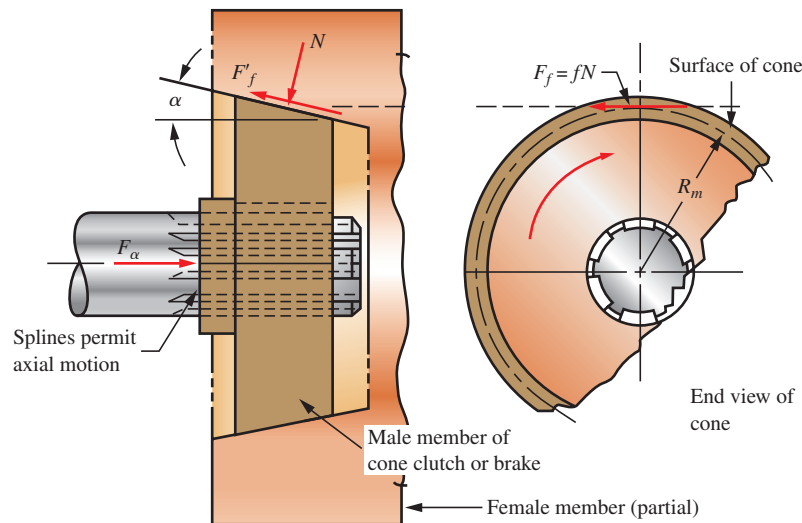


FIGURE 22-16 Cone clutch or brake

opposes the tendency for the member having the internal cone surface to move axially away from the external cone. We will call this force F'_f and compute it also from

$$F'_f = fN$$

For the equilibrium condition of the external cone, the sum of the horizontal forces must be zero. Then

$$F_a = N \sin \alpha + F'_f \cos \alpha = N \sin \alpha + fN \cos \alpha$$

$$F_a = N(\sin \alpha + f \cos \alpha)$$

or

$$N = \frac{F_a}{\sin \alpha + f \cos \alpha} \quad (22-12)$$

Substituting this into Equation (22-8) gives

➤ Frictional Torque on a Cone Clutch or Brake

$$T_f = \frac{fR_m F_a}{\sin \alpha + f \cos \alpha} \quad (22-13)$$

Example Problem 22-7

Compute the axial force required for a cone brake if it is to exert a braking torque of 50 lb·ft. The mean radius of the cone is 5.0 in. Use $f = 0.25$. Try cone angles of 10°, 12°, and 15°.

Solution We can solve Equation (22-13) for the axial force F_a :

$$F_a = \frac{T_f(\sin \alpha + f \cos \alpha)}{fR_m} = \frac{(50 \text{ lb} \cdot \text{ft})(\sin \alpha + 0.25 \cos \alpha)}{(0.25)(5.0/12) \text{ ft}}$$

$$F_a = 480(\sin \alpha + 0.25 \cos \alpha) \text{ lb}$$

Then the values of F_a as a function of the cone angle are as follows:

$$\text{For } \alpha = 10^\circ, F_a = 202 \text{ lb}$$

$$\text{For } \alpha = 12^\circ, F_a = 217 \text{ lb}$$

$$\text{For } \alpha = 15^\circ, F_a = 240 \text{ lb}$$

22-14 DRUM BRAKES

Short Shoe Drum Brakes

Figure 22-17 shows a sketch of a drum brake in which the actuating force W acts on the lever that pivots on pin A. This creates a normal force between the shoe and the

rotating drum. The resulting friction force is assumed to act tangential to the drum if the shoe is short. The friction force times the radius of the drum is the friction torque that slows the drum.

The objectives of the analysis are to determine the relationship between the applied load and the friction

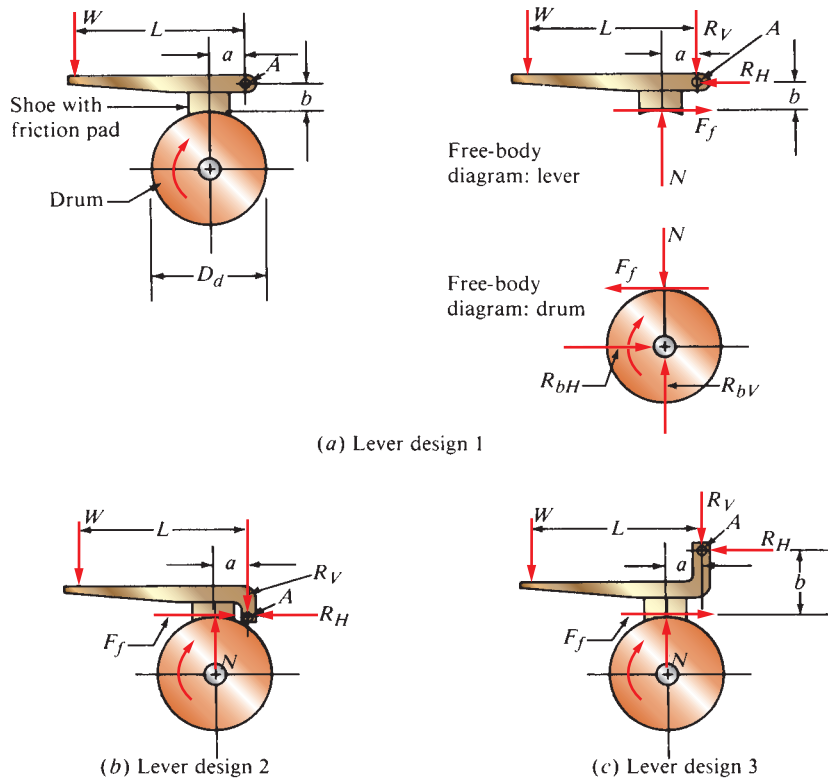


FIGURE 22-17 Short shoe drum brake

force and to be able to evaluate the effect of design decisions such as the size of the drum, the lever dimensions, and the placement of the pivot A. The free-body diagrams in Figure 22-17(a) support this analysis. For the lever, we can sum moments about the pivot A:

$$\Sigma M_A = 0 = WL - Na + F_f b \quad (22-14)$$

But note that $F_f = fN$ or $N = F_f/f$; where f = coefficient of friction. Then

$$0 = WL - F_f a/f + F_f b = WL - F_f(a/f - b)$$

Solving for W gives

$$W = \frac{F_f(a/f - b)}{L} \quad (22-15)$$

Solving for F_f gives

Friction Force on Drum Brake

$$F_f = \frac{WL}{(a/f - b)} \quad (22-16)$$

We can use these equations for the friction torque by noting

Friction Torque

$$T_f = F_f D_d / 2 \quad (22-17)$$

where D_d = diameter of the drum

Note the alternate positions of the pivot in Parts (b) and (c) of Figure 22-17. In (b), the dimension $b = 0$.

Example Problem 22-8

Compute the actuation force required for the short shoe drum brake of Figure 22-17(a) to produce a friction torque of 50 lb·ft. Use a drum diameter of 10.0 in, $a = 3.0$ in, and $L = 15.0$ in. Consider values of f of 0.25, 0.50, and 0.75, and different points of location of pivot A such that b ranges from 0 to 6.0 in.

Solution The required friction force can be found from Equation (22-17):

$$F_f = 2T_f/D_d = (2)(50 \text{ lb} \cdot \text{ft})/(10/12 \text{ ft}) = 120 \text{ lb}$$

In Equation (22-15), we can substitute for a , L , and F_f :

$$W = \frac{F_f(a/f - b)}{L} = \frac{120 \text{ lb}[(3.0 \text{ in})/f - b]}{15.0 \text{ in}} = 8(3/f - b) \text{ lb}$$

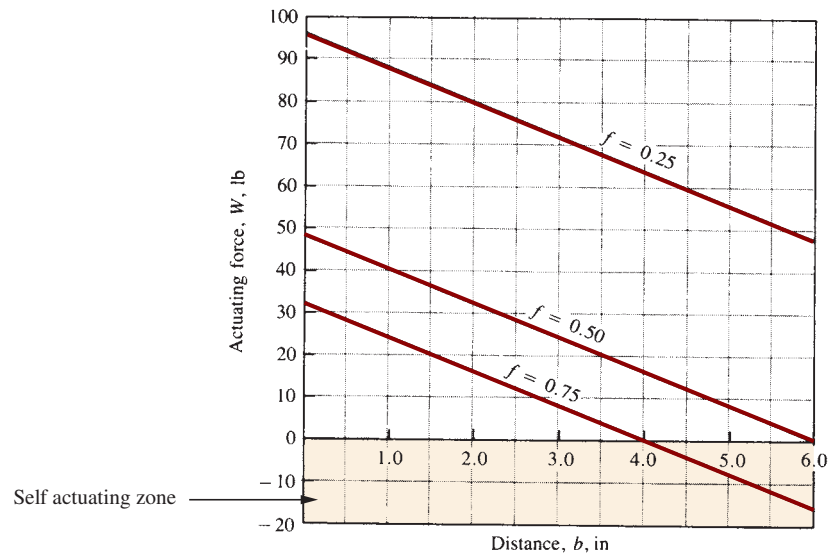


FIGURE 22-18 Results: actuating load force versus distance b

We can substitute the varying values of f and b into this last equation to compute the data for the curves of Figure 22-18, showing the actuating force versus the distance b for different values of f . Note that for some combinations, the value of W is *negative*. This means that the brake is *self-actuating* and that an upward force on the lever would be required to release the brake.

Long Shoe Drum Brakes

The assumption used for short shoe brakes, that the resultant friction force acted at the middle of the shoe, cannot be used in the case of shoes covering more than about 45° of the drum. In such cases, the pressure between the friction lining and the drum is very nonuniform, as is the moment of the friction force and of the normal force with respect to the pivot of the shoe.

The following equations govern the performance of a long shoe brake, using the terminology from Figure 22-19. (See Reference 4.)

1. Friction torque on drum:

$$T_f = r^2 f w p_{\max} (\cos \theta_1 - \cos \theta_2) \quad (22-18)$$

2. Actuation force:

$$W = (M_N + M_f) / L \quad (22-19)$$

where M_N = moment of normal force with respect to the hinge pin

$$M_N = 0.25 p_{\max} w r C [2(\theta_2 - \theta_1) - \sin 2\theta_2 + \sin 2\theta_1] \quad (22-20)$$

M_f = moment of friction force with respect to the hinge pin

$$M_f = f p_{\max} w r [r(\cos \theta_1 - \cos \theta_2) + 0.25 C (\cos 2\theta_2 - \cos 2\theta_1)] \quad (22-21)$$

The sign of M_f is negative (–) if the drum surface is moving away from the pivot and positive (+) if it is moving toward the pivot.

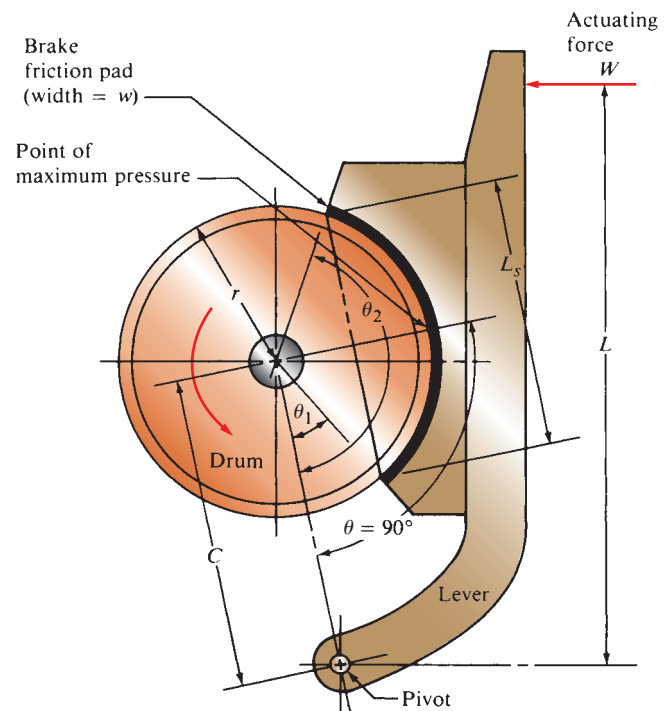


FIGURE 22-19 Terminology for long shoe drum brake

3. Friction power:

$$P_f = T_f n / 63\,000 \text{ hp} \quad (22-22)$$

where n = rotational speed in rpm

4. Brake shoe area (Note: Projected area is used):

$$A = L_s w = 2wr \sin[(\theta_2 - \theta_1)/2] \quad (22-23)$$

5. Wear rating:

$$WR = P/A \quad (22-24)$$

The use of these relationships in the design and analysis of a long shoe brake is shown in Example Problem 22-9.

Example Problem 22-9

Design a long shoe drum brake to produce a friction torque of 750 lb·in to stop a drum from 120 rpm.

Solution

Step 1. Select a brake friction material, and specify the desired maximum pressure and the design value for the coefficient of friction. Table 22-2 lists some general properties for friction materials. Actual test values or specific manufacturer's data should be used where possible. The design value for $p_{\max}$ should be far less than the permissible pressure listed in Table 22-2 in order to improve wear life.

For this problem, let's choose a woven fabric material and design for approximately 75 psi maximum pressure. Note, as shown in Figure 22-19, that the maximum pressure occurs at the section 90° from the pivot. If the shoe does not extend at least 90°, the equations used here are not valid. (See Reference 4.) Also, let's use $f = 0.25$ for the design.

Step 2. Propose trial values for the geometry of the brake drum and the brake pad. Several design decisions must be made here. The general arrangement shown in Figure 22-19 can be used as a guide. But your specific application and creativity may lead you to modify the arrangement.

Trial values are $r = 4.0$ in; $C = 8.0$ in; $L = 15$ in; $\theta_1 = 30^\circ$; and $\theta_2 = 150^\circ$.

Step 3. Solve for the required width of the shoe from Equation (22-18):

$$w = \frac{T_f}{r^2 f p_{\max} (\cos \theta_1 - \cos \theta_2)}$$

For this problem,

$$w = \frac{750 \text{ lb} \cdot \text{in}}{(4.0 \text{ in})^2 (0.25) (75 \text{ lb/in}^2) (\cos 30^\circ - \cos 150^\circ)} = 1.443 \text{ in}$$

For convenience, let $w = 1.50$ in. Because the maximum pressure is inversely proportional to the width, the actual maximum pressure will be

$$p_{\max} = 75 \text{ psi} (1.44/1.50) = 72.17 \text{ psi}$$

Step 4. Compute M_N from Equation (22-20). The value of $\theta_2 - \theta_1$ must be in radians, with π radians = 180°. Then

$$\theta_2 - \theta_1 = 120^\circ (\pi \text{ rad}/180^\circ) = 2.0944 \text{ rad}$$

The moment of the normal force on the shoe is

$$\begin{aligned} M_N &= 0.25 (72.17 \text{ lb/in}^2) (1.50 \text{ in}) (4.0 \text{ in}) (8.0 \text{ in}) \\ &\quad [2(2.09) - \sin(300^\circ) + \sin(60^\circ)] \\ M_N &= 5128 \text{ lb} \cdot \text{in} \end{aligned}$$

Step 5. Compute the moment of the friction force on the shoe, M_f , from Equation (22-21):

$$\begin{aligned} M_f &= 0.25 (72.17 \text{ lb/in}^2) (1.50 \text{ in}) (4.0 \text{ in}) \\ &\quad [(4.0 \text{ in}) (\cos 30^\circ - \cos 150^\circ) \\ &\quad + 0.25 (8.0 \text{ in}) (\cos 300^\circ - \cos 60^\circ)] \\ M_f &= 749.8 \text{ lb} \cdot \text{in} \end{aligned}$$

Step 6. Compute the required actuation force, W , from Equation (22-19):

$$W = (M_N - M_f)/L = (5128 - 749.8)/(15) = 291.8 \text{ lb}$$

Note the minus sign for M_f because the drum surface is moving away from the pivot.

Step 7. Compute the frictional power from Equation (22-22):

$$P_f = T_f n / (63\,000) = (749.8)(120) / (63\,000) = 1.428 \text{ hp}$$

Step 8. Compute the projected area of the brake shoe from Equation (22-23):

$$A = L_s w = 2wr \sin[(\theta_2 - \theta_1)/2]$$

$$A = 2(1.50 \text{ in})(4.0 \text{ in}) \sin(120^\circ/2) = 10.392 \text{ in}^2$$

Step 9. Compute the wear ratio, WR :

$$WR = P_f / A = 1.428 \text{ hp} / 10.392 \text{ in}^2 = 0.137 \text{ hp/in}^2$$

Step 10. Evaluate the suitability of the results. In this problem, we would need more information about the application to evaluate the results. However, the wear rating seems reasonable for average service (see Section 22-11), and the geometry seems acceptable.

22-15 BAND BRAKES

Figure 22-20 shows the typical configuration of a *band brake*. The flexible band, usually made of steel, is faced with a friction material that can conform to the curvature of the drum. The application of a force to the lever puts tension in the band and forces the friction material against the drum. The normal force, thus created, causes the friction force tangential to the drum surface to be created, retarding the drum.

The tension in the band decreases from the value P_1 at the pivot side of the band to P_2 at the lever side. The net torque on the drum is then

$$T_f = (P_1 - P_2)r \quad (22-25)$$

where r = radius of the drum

The relationship between P_1 and P_2 can be shown (see Reference 4) to be the logarithmic function

$$P_2 = P_1 / e^{f\theta} \quad (22-26)$$

where θ = total angle of coverage of the band in radians

The point of maximum pressure on the friction material occurs at the end nearest the highest tension, P_1 , where

$$P_1 = p_{\max} r w \quad (22-27)$$

and w is the width of the band.

For the two types of band brakes shown in Figure 22-20, the free-body diagrams of the levers can be used to show

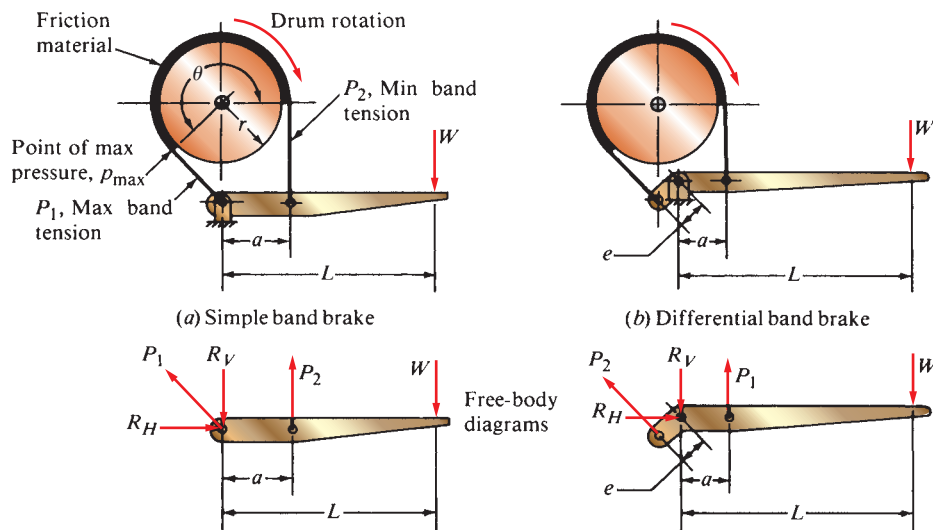


FIGURE 22-20 Band brake design

the following relationships for the actuating force, W , as a function of the tensions in the band. For the simple band brake of Figure 22–20(a),

$$W = P_2 a / L \quad (22-28)$$

The style shown in Figure 22–20(b) is called a *differential band brake*, where the actuation force is

$$W = (P_2 a - P_1 e) / L \quad (22-29)$$

The design procedure is presented in Example Problem 22–10.

Example Problem 22–10

Design a band brake to exert a braking torque of at least 720 lb·in while slowing the drum from 120 rpm.

Solution

Step 1. Select a material and specify a design value for the maximum pressure. A woven friction material is desirable to facilitate the conformance to the cylindrical drum shape. Let's use $p_{\max} = 25$ psi and a design value of $f = 0.25$. See Section 22–10.

Step 2. Specify a trial geometry: r , θ , w . For this problem, let's try $r = 6.0$ in, $\theta = 225^\circ$, and $w = 2.0$ in. Note that $225^\circ = 3.93$ rad.

Step 3. Compute the maximum band tension, P_1 , from Equation (22–27):

$$P_1 = p_{\max} r w = (25 \text{ lb/in}^2)(6.0 \text{ in})(2.0 \text{ in}) = 300 \text{ lb}$$

Step 4. Compute tension P_2 from Equation (22–26):

$$P_2 = \frac{P_1}{e^{\theta}} = \frac{300 \text{ lb}}{e^{(0.25)(3.93)}} = 112.3 \text{ lb}$$

Step 5. Compute the friction torque, T_f :

$$T_f = (P_1 - P_2)r = (300 - 112.3)(6.0) = 1126 \text{ lb·in}$$

Note: Repeat Steps 2–5 until you achieve a satisfactory geometry and friction torque. Let's try a smaller design, say, $r = 5.0$ in:

$$P_1 = (25)(5.0)(2.0) = 250 \text{ lb}$$

$$P_2 = \frac{250 \text{ lb}}{e^{(0.25)(3.93)}} = 93.59 \text{ lb}$$

$$T_f = (250 - 93.59)(5.0) = 782.0 \text{ lb·in (okay)}$$

Step 6. Specify the geometry of the lever, and compute the required actuation force. Let's use $a = 5.0$ in and $L = 15.0$ in. Then

$$W = P_2(a/L) = 93.59 \text{ lb}(5.0/15.0) = 31.20 \text{ lb}$$

Step 7. Compute the average wear rating from $WR = P_f A$:

$$A = 2\pi r w (\theta/360) = 2(\pi)(5.0 \text{ in})(2.0 \text{ in})(225/360) = 39.27 \text{ in}^2$$

$$P_f = T_f n / (63\,000) = (782.0)(120) / (63\,000) = 1.490 \text{ hp}$$

$$WR = P_f / A = (1.490 \text{ hp}) / (39.27 \text{ in}^2) = 0.0379 \text{ hp/in}^2$$

This should be conservative for average service.

22–16 OTHER TYPES OF CLUTCHES AND BRAKES

The chapter thus far has concentrated on clutches and brakes using friction materials to transmit the torque between rotating members, but many other types are available. Brief descriptions are given below, but specific design information is not given. Most are unique to the manufacturer, and the application data are available through catalogs.

Jaw Clutch

The teeth of the mating sets of jaws are brought into engagement by sliding one or both members axially. The teeth may be straight-sided or triangular, or they may incorporate some smooth curve to facilitate engagement. Once the teeth are engaged, there is a positive transmission of torque. The jaw clutch is normally engaged while the system is stopped or is running very slowly.

Ratchet

Although not strictly a clutch, the familiar ratchet and pawl permits alternate engagement and disengagement of moving members, and thus it can be used in similar applications. Typically, the ratchet moves only a small fraction of a revolution per cycle.

Sprag, Roller, and Cam Clutches

There are differences in the specific geometry of sprag, roller, and cam clutches, but they all perform similar functions. When the input shaft is rotating in the driving direction, the internal elements (sprags, rollers, or cams) are wedged between the driving and driven members and thus transmit torque. But when the input member rotates in the opposite direction, the internal elements move out of engagement, and no torque is transmitted. Thus, they can be used for applications similar to ratchets, but with much smoother operation and with a virtually infinite amount of incremental motion. Another application is *backstopping*, in which the clutch runs free when the machine is being driven in the intended direction. But if the drive is disengaged and the load starts to reverse direction, the clutch locks up and prevents motion. This type of clutch is also used for *overrunning*: a positive drive as long as the load rotates no faster than the driver. If the load tends to run faster (overrun) than the driver, the clutch elements disengage. This protects the equipment that might be damaged due to overspeeding. (See Internet site 6.)

Fiber Clutch

A fiber clutch operates in a manner similar to the overrunning clutches described previously. But instead of driving through solid elements, the torque is transmitted through stiff fibers that have a preferred orientation. When rotated in a direction opposite the preferred direction, the fibers “lie down,” and no torque is transmitted.

Wrap-Spring Clutch

Again used in cases similar to the overrunning clutches, the *wrap-spring clutch* is made from a rectangular wire and normally has an inside diameter slightly larger than that of the shaft on which it is installed. Thus, no torque is transmitted. But when one end of the spring is restrained, the spring “wraps down” tightly on the surface of the shaft, and torque is transmitted positively through the spring. (See Internet site 4.)

Single-Revolution Clutches

It is frequently desired to have a machine cycle one complete revolution and then come to a stop. The *single-revolution clutch* provides this feature. After it is tripped,

it drives the output shaft until reaching a positive stop at the end of one revolution. Some types can be engaged for more than one revolution, but they will return to a fixed position, for example, at the top of the stroke of a press. (See Internet site 6.)

Fluid Clutches

The *fluid clutch* is made up of two separate parts with no mechanical connection between them. A fluid fills a cavity between the parts, and as one member rotates, it tends to shear the fluid, causing torque to be transmitted to the mating element. The resulting drive is smooth and soft because load peaks will simply cause one member to move relative to the other. In this situation, it is similar to the slip clutch described earlier.

Eddy Current Drive

When a conducting disc moves through a magnetic field, *eddy currents* are induced in the disc, causing a force to be exerted on the disc in a direction opposite to the direction of rotation. The force can be used to brake the disc or to transmit torque to a mating part, such as a clutch. An advantage of this type of unit is that there is no mechanical connection between the elements. The torque can be controlled by varying the current to the electromagnets.

Overload Clutches

The drive is positive, provided that the torque is below some set value. At higher torques, some element is disengaged automatically. One type uses a series of spherical balls positioned in detents and held under a spring force. When the tripping torque level is reached, the balls are forced out of the detents and disengage the drive. (See Internet site 6.)

Tensioners

The production of continuous products such as wire, webs of paper, or plastic film require that the drive system be carefully controlled to maintain a light tension without breaking the product. Similar control must be exerted when winding coils of paper, foil, or sheet metal as they are produced, or when unwinding them to feed a process such as printing or press forming. Drives for cranes and hoists must provide controlled braking while loads are lowered. In such cases, brakes are required to exert some braking action while allowing smooth motion. Many of the brake designs reviewed in this chapter can accomplish this function by moderating the force applied between the friction elements. An example is the air-actuated brake shown in Figure 22–6. The braking torque depends on the air pressure applied that can be controlled by an operator or automatically. (See also Internet site 5.)

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2. Juvinall, Robert C., and Kurt M. Marshek. *Fundamentals of Machine Component Design*. 5th ed. New York: John Wiley & Sons, 2012.
3. Orthwein, William C. *Clutches and Brakes: Design and Selection*. 2nd ed. Boca Raton, FL: CRC Press, 2004.
4. Budynas, R. G., and K. J. Nisbett. *Shigley's Mechanical Engineering Design*. 10th ed. New York: McGraw-Hill, 2015.
5. SAE International. *Standard J286 Clutch Friction Test Machine Guidelines*. Warrendale, PA: Society of Automotive Engineers, 2012.
6. SAE International. *Standard J661 Brake Lining Quality Control Test Procedure*. Warrendale, PA: Society of Automotive Engineers, 2012.
7. SAE International. *Standard J866 Friction Coefficient Identification System for Brake Linings*. Warrendale, PA: Society of Automotive Engineers, 2012.
8. Hilliard Corporation. Manufacturer of a wide variety of clutches and brakes for industry and commercial equipment applications. Included are centrifugal clutches, single-revolution clutches, overrunning clutches, ball detent overload release clutches, slip clutch torque limiters, intermittent drives, and caliper disc brakes. From the home page, select *Motion Control*.
9. Tol-O-Matic, Inc. Manufacturer of pneumatically and hydraulically operated caliper disc brakes, disc cone clutches, and other automation and motion control products. From the home page select *Products*, then *Power Transmission*.
10. Electroid Company. Manufacturer of a wide variety of clutches, brakes, and tensioners for industrial, commercial, and aerospace applications. From the home page select *Brakes & Clutches*.
11. Regal Beloit Americas, Inc. Manufacturer of various styles of clutches and torque overload devices under the Morse and Browning brands. Select *Products*, then *Drive Components*, then *Mechanical CAM Clutches or Torque Overload Devices*. Site also includes a variety of technical literature under the *Resources* tab.

INTERNET SITES FOR CLUTCHES AND BRAKES

1. GKN Land Systems. Manufacturer of clutches, disc brakes, driveshafts, and other powertrain components for the automotive, truck, and off-road equipment markets. Brand names include GKN, Rockford® Clutch, Mechanics® Driveshafts, Stromag, Walterscheid clutches, Rotowell brakes, and others.
2. Baldor/Dodge. Manufacturer of Dodge brakes and clutches for industrial machinery, conveying equipment, and textile industry. Select *Products*, then *Couplings, Clutches, & Brakes*. Also offers a wide array of gear drives, belt drives, bearings, and numerous other power transmission products.
3. BorgWarner, Inc. Manufacturer of automotive transmission clutch sets and other components. Select *Products*, then *Transmission Systems*.
4. Warner Electric, Inc. Manufacturer of clutch and brake systems for industrial, turf and garden, wind turbine, and vehicular applications. Includes a wide variety of disk type, wrap spring, and magnetic particle clutches and brakes for start, stop, hold, and tension control. Catalog information can be downloaded from the Literature section. A member of Altra Industrial Motion Group.
5. Eaton/Airflex. Manufacturer of industrial clutches and brakes using the expanding tube principle actuated by pneumatic or hydraulic pressure. Applications include engines, paper making machinery, power presses, brakes, shears, marine drives, well drilling, and tensioning, winding, and unwinding equipment. Online product data and descriptions, product application information, and technical information. From the home page, select *Products & Solutions*, then *Industrial Clutches and Brakes*.

PROBLEMS

1. Specify the required torque rating for a clutch to be attached to a motor shaft running at 1750 rpm. The motor is rated at 5.0 hp and is of the design B type.
2. Specify the required torque rating for a clutch to be attached to a diesel engine shaft running at 2500 rpm. The engine is rated at 75.0 hp.
3. Specify the required torque rating for a clutch to be attached to an electric motor shaft running at 1150 rpm. The motor is rated at 0.50 hp and drives a light fan.
4. An alternative design for the system described in Problem 1 is being considered. Instead of putting the clutch on the motor shaft, it is desired to place it on the output shaft of a speed reducer that rotates at 180 rpm. The power transmission is still approximately 5.0 hp. Specify the required torque rating for the clutch.
5. Specify the required brake torque rating for each of the conditions in Problems 1–4 for average industrial conditions.
6. Specify the required torque rating for a clutch in $\text{N} \cdot \text{m}$ if it is attached to a design B electric motor shaft rated at 20.0 kW and rotating at 3450 rpm.
7. A clutch-brake module is to be connected between a design C electric motor and a speed reducer. The motor is rated at 50.0 kW at 900 rpm. Specify the required torque ratings for the clutch and the brake portions of the module for average industrial service. The drive is to a large conveyor.
8. Compute the torque required to accelerate a solid steel disc from rest to 550 rpm in 2.0 s. The disc has a diameter of 24.0 in and is 2.5 in thick.
9. The assembly shown in Figure P22–9 is to be stopped by a brake from 775 rpm to zero in 0.50 s or less. Compute the required brake torque.
10. Compute the required clutch torque to accelerate the system shown in Figure P22–10 from rest to a motor speed of 1750 rpm in 1.50 s. Neglect the inertia of the clutch.

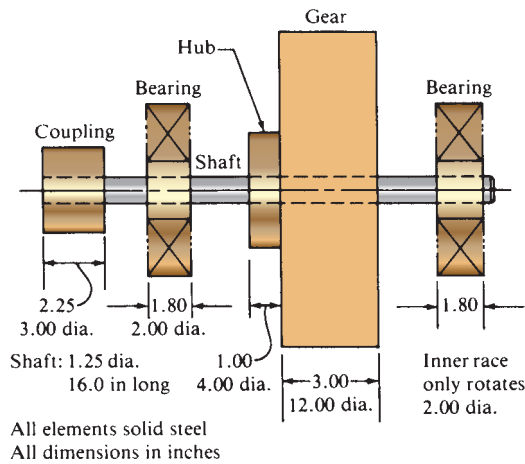


FIGURE P22-9 (Problems 9, 14, and 16)

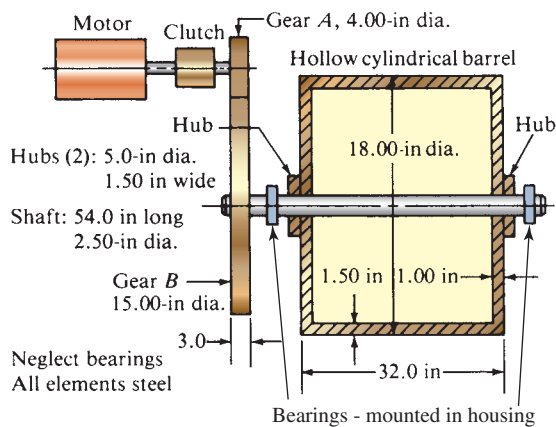


FIGURE P22-10

11. A winch, sketched in Figure P22-11, is lowering a load at the speed of 50 ft/min. Compute the required torque rating for the brake on the winch shaft to stop the system in 0.25 s.

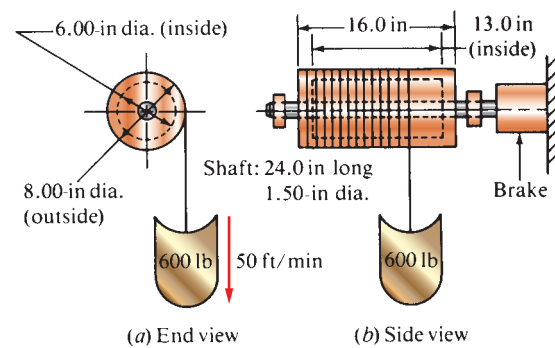


FIGURE P22-11

12. Figure P22-12 shows a tumbling barrel being driven through a wormgear reduction unit. Evaluate the torque rating required for a clutch to accelerate the barrel to 38.0 rpm from rest in 2.0 s (a) if the clutch is placed on the motor shaft and (b) if it is placed at the output of the reducer. Neglect the inertia of the gear shafts, the bearing races, and the clutch. Consider the worm and wormgear to be solid cylinders.
13. Compute the dimensions of an annular plate-type brake to produce a braking torque of 75 lb·in. Air pressure will develop a normal force of 150 lb between the friction surfaces. Use a coefficient of friction of 0.25. The brake will be used in average industrial service, stopping a load from 1150 rpm.
14. Design a plate-type brake for the application described in Problem 9. Specify the design coefficient of friction, the dimensions of the plate, and the axial force required.
15. Compute the axial force required for a cone clutch if it is to exert a driving torque of 15 lb·ft. The cone surface has a mean diameter of 6.0 in and an angle of 12° . Use $f = 0.25$.
16. Design a cone brake for the application described in Problem 9. Specify the design coefficient of friction, the mean diameter of the cone surface, and the axial force required.

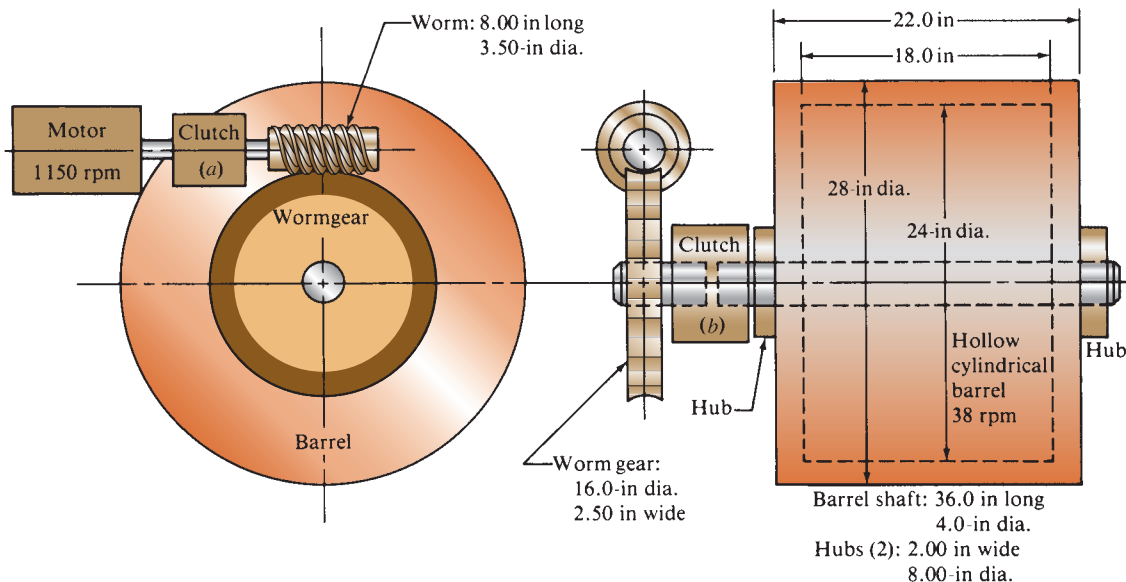


FIGURE P22-12

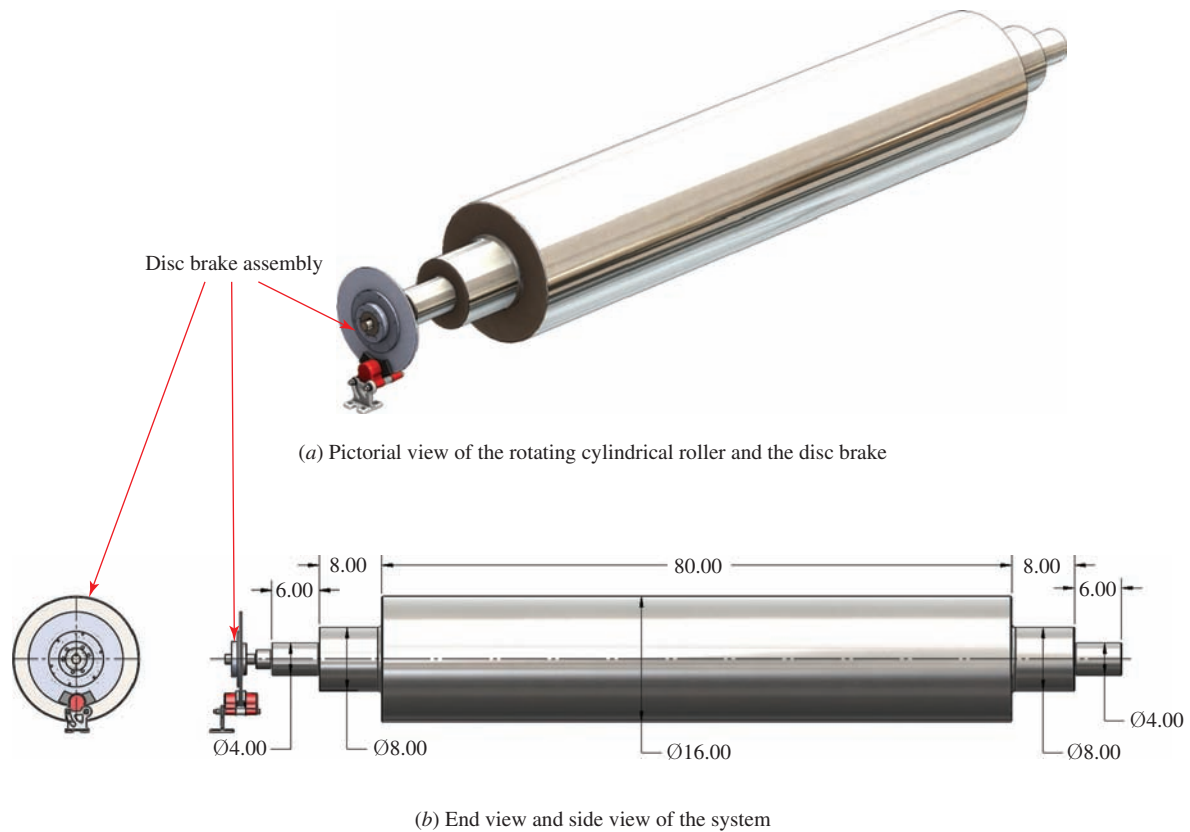


FIGURE P22-22

17. Compute the actuation force required for the short shoe drum brake of Figure 22-17(a) to produce a friction torque of $150 \text{ lb} \cdot \text{ft}$. Use a drum diameter of 12.0 in , $a = 4.0 \text{ in}$, and $L = 24.0 \text{ in}$. Use $f = 0.25$ and $b = 5.0 \text{ in}$.
18. For all other data from Problem 17 being the same, determine the required dimension b for the brake to be self-actuating.
19. Design a short shoe drum brake to produce a torque of $100 \text{ lb} \cdot \text{ft}$. Specify the drum diameter, the configuration of the actuation lever, and the actuation force.
20. Design a long shoe drum brake to produce a friction torque of $100 \text{ lb} \cdot \text{ft}$ to stop a load from 480 rpm . Specify the friction material, the drum size, the shoe configuration, pivot locations, and actuation force.
21. Design a band brake to exert a braking torque of $75 \text{ lb} \cdot \text{ft}$ while slowing a drum from 350 rpm to rest. Specify a material, the drum diameter, the band width, the angle of coverage of the friction material, the actuation lever configuration, and the actuation force.
22. The solid steel cylinder shown in Figure P22-22 is used as a guide roller for a paper processing system. In normal operation, it rotates at an angular velocity of 800 rpm . A disc caliper brake is mounted on the left end of the roller to stop it when the system is shut down. When the brake is applied, the roller must stop in a time of 5.0 seconds or less. The magnitude of the torque applied to the roller is required to select the proper size for the disc caliper brake. Solve for the following:
 - (a) Mass moment of inertia, I , of the cylinder
 - (b) The required deceleration rate for the cylinder
 - (c) The braking torque required to stop the cylinder in 5.0 seconds or less.

DESIGN PROJECTS

23-1 Objectives of This Chapter

23-2 Design Projects

23-1 OBJECTIVES OF THIS CHAPTER

One of the primary focuses of this book has been to emphasize the integration of machine elements into complete mechanical designs. The interfaces between machine elements have been discussed for many examples. The forces exerted on one element by another have been computed. Commercially available components and complete devices have been shown in figures throughout the book.

Although these discussions and examples are helpful, one of the better ways to learn mechanical design is to *do* mechanical design. You should complete these steps:

- Define the detailed functions and design requirements for the design.
- Conceptualize several approaches to a design.
- Decide which approach to complete.
- Complete the design of each element in detail.
- Make assembly and detail drawings to communicate your design to others who may use it or be responsible for its fabrication.
- Specify completely the purchased components that are part of the design.

Following are several projects that call for you to do these operations. You or your instructor may modify or amplify the projects to suit individual needs or the available time or information. As in most design projects, many solutions are possible. Different solutions from several members of a class could be compared and critiqued to enhance learning. It may be helpful at this time to review Sections 1-4 and 1-5 about the functions and design requirements for design and a philosophy of design. Also, the problems at the end of Chapter 1 asked that you write a set of functions and design requirements for several of the same devices. If you have already done so, they can be used as a part of these exercises.

23-2 DESIGN PROJECTS

Automobile Hood Latch

Design a hood latch for an automobile. The latch must be able to hold the hood securely closed during operation of the vehicle. But it should be easy to open for servicing the contents of the engine compartment. Theft-proofing is an important design goal. Attachment of the latch to the frame of the car and to the hood should be defined. Mass production should be a requirement.

Hydraulic Lift

Design a hydraulic lift to be used for car repair. Obtain pertinent dimensions from representative cars for initial height, extended height, design of the pads that contact the car, and so on. The lift will raise the entire car.

Car Jack

Design a floor jack for a car to lift either the entire front end or the entire rear end. The jack may be powered by hand, using mechanical or hydraulic actuation. Or it may be powered by pneumatic pressure or electrical power.

Portable Crane

Design a portable crane to be used in homes, small industries, warehouses, and garages. It should have a capacity of at least 1000 lb (4.45 kN). Typical uses would be to remove an engine from a car, lift machine components, or load trucks.

Can Crusher

Design a machine to crush soft-drink or beer cans. The crusher would be used in homes or restaurants as an aid to recycling efforts. It could be operated either by hand or electrically. It should crush the cans to approximately 20% of their original volume.

Transfer Device

Design an automatic transfer device for a production line. The parts to be handled are steel castings with the following characteristics:

Weight: 42.0 lb (187 N)

Size: Cylindrical; 6.75-in diameter and 10.0 in high. Exterior surface is free of projections or holes and has a reasonably smooth, as-cast finish.

Transfer rate: Continuous flow, 2.00 s between parts.

Parts enter at a 24.0-in elevation on a roller conveyor. They must be elevated to 48.0 inches in a space of 60.0 in horizontally. They leave on a separate conveyor.

Drum Dumper

Design a drum dumper. The machine is to raise a 55-gal drum of bulk material from floor level to a height of 60.0 in and dump the contents of the drum into a hopper.

Paper Feeder

Design a paper feed device for a copier. The paper must be fed at a rate of 120 sheets per min.

Gravel Conveyor

Design a conveyor to elevate gravel into a truck. The top edge of the truck bed is 8.0 ft (2.44 m) off the ground. The bed is 6.5 ft wide, 12.0 ft long, and 4.0 ft deep (1.98 m $\times$ 3.66 m $\times$ 1.22 m). It is desired to fill the truck in 5.0 min or less.

Construction Lift

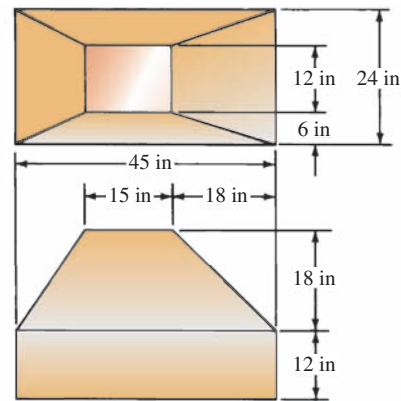
Design a construction lift. The lift will raise building materials from ground level to any height up to 40.0 ft (12.2 m). The lift will be at the top of a rigid scaffold that is not a part of the design project. It will raise a load of up to 500 lb (2.22 kN) at the rate of 1.0 ft/s (0.30 m/s). The load will be on a pallet, 3.0 ft by 4.0 ft (0.91 m $\times$ 1.22 m). At the top of the lift, means must be provided to bring the load onto a platform that supports the lift.

Packaging Machine

Design a packaging machine. Toothpaste tubes are to be taken from a continuous belt and inserted into cartons. Any standard tube size may be chosen. The device may include the means to close the cartons after the tube is in place.

Carton Packer

Design a machine to insert 24 cartons of toothpaste into a shipping case.



Weld all seams with weld line in horizontal plane

FIGURE 23-1 Frame to be handled by a weld positioner

Robot Gripper

Design a gripper for a robot to grasp a spare-tire assembly from a rack and insert it into the trunk of an automobile on an assembly line. Obtain dimensions from a particular car.

Weld Positioner

Design a weld positioner. A heavy frame is made of welded steel plate in the shape shown in Figure 23-1. The welding unit will be robot-guided, but it is essential that the weld line be horizontal as the weld proceeds. Design the device to hold the frame securely and move it to present the part to the robot. The plate has a thickness of 3/8 in (9.53 mm).

Garage Door Opener

Design a garage door opener.

Spur Gear Speed Reducer, Single-Reduction

Design a complete single-reduction, spur gear-type speed reducer. Specify the two gears, two shafts, four bearings, and a housing. Use any of the data from Chapter 9, Problems 60–70.

Spur Gear Speed Reducer, Double-Reduction

Design a complete double-reduction, spur gear-type speed reducer. Specify the four gears, three shafts, six bearings, and a housing. Use any of the data from Chapter 9, Problems 74–76.

Helical Gear Speed Reducer, Single-Reduction

Design a complete single-reduction, helical gear-type speed reducer. Use any of the data from Chapter 10, Problems 5–11.

Bevel Gear Speed Reducer, Single-Reduction

Design a complete single-reduction, bevel gear-type speed reducer. Use any of the data from Chapter 10, Problems 14–17.

Wormgear Speed Reducer, Single-Reduction

Design a complete single-reduction, wormgear-type speed reducer. Use any of the data from Chapter 10, Problems 18–24.

Lift Device Using Acme Screws

Design a device similar to that sketched in Figure 17–4. An electric motor drives the worm at a speed of 1750 rpm. The two Acme screws rotate and lift the yoke, which in turn lifts the hatch. See Example Problem 17–1 for additional details. Complete the entire unit, including the wormgear set, the chain drive, the Acme screws, the bearings, and their mountings. The hatch is 48 in wide and 155 in long and its weight is 25 000 lb. The screws should be nominally 30 in (762 mm) long. The total motion of the yoke will be 24 in, to be completed in 15.0 s or less.

Lift Device Using Ball Screws

Repeat the design of the lift device, using ball screws instead of Acme screws.

Brake for a Drive Shaft

Design a brake. A rotating load (as sketched in Figure P22–9) is to be stopped from 775 rpm in 0.50 s or less. Use any type of brake described in Chapter 22, and complete the design details, including the actuating means: springs, air pressure, manual lever, and so on. Show the brake attached to the shaft of Figure P22–9.

Brake for a Winch

Design a complete brake for the application shown in Figure P22–11 and described in Problem 11 in Chapter 22.

Indexing Drive

Design an indexing drive for an automatic assembly system. The items to be moved are mounted on a square steel fixture plate, 6.0 in (152 mm) on a side and 0.50 in (12.7 mm) thick. The total weight of each assembly is 10.0 lb (44.5 N). The center of each fixture (intersection of its diagonals) is to move 12.0 in (305 mm) with each index. The index is to be completed in 1.0 s or less, and the fixture must be held stationary at each station for a minimum of 2.0 s. Four assembly stations are required. The arrangement may be linear, rotary, or any other, provided that the fixtures move in a horizontal plane.

Child's Ferris Wheel

Design a child's Ferris wheel. It should be capable of holding one to four children weighing up to 80 lb (356 N) each. The rotational speed should be 1 rev in 6.0 s. It should be driven by an electric motor.

Merry-Go-Round

Design an amusement ride for small children, age 6 or younger, that they can sit in safely while enjoying an interesting motion pattern. At least two children at a time can ride the machine. The ride will be marketed to shopping centers and department stores to amuse customers' children.

Backyard Amusement Ride

Design a backyard amusement ride in which small coaster wagons are pulled along a circular path. The ride should be powered by an electric motor. Each wagon is 1.0 m long (39.4 in) and 0.50 m wide (19.7 in). The four wheels are 150 mm (6.0 in) in diameter. The wagon is to be attached to the drive rod at the point where the handle would normally be. The radial distance to the attachment point is to be 2.0 m (6.6 ft). The wagons are to make 1 rev in 8.0 s. Provide a means of starting and stopping the drive.

Transfer Device

Design a device to move automotive camshafts between processing stations. Each movement is to be 9.0 in (229 mm). The camshaft is to be supported on two unfinished bearing surfaces having a diameter of 3.80 in (96.5 mm) and an axial length of 0.75 in (19.0 mm). The spread between the bearing surfaces is 15.75 in (400.0 mm). Each camshaft weighs 16.3 lb (72.5 N). One motion cycle is to be completed each 2.50 s. Design the complete mechanism, including the drive from an electric motor.

Chain Conveyor

Design a powered, straight chain conveyor to move eight pallets along an assembly line. The pallets are 18 in long and 12 in wide. The maximum weight of each pallet and the product it carries is 125 lb. At the end of the conveyor, an external downward force of 500 lb is applied to the product which must be carried through the pallet to the conveyor structure. You may design the configuration of the sides and bottom of the pallet.

"You Are the Designer" Projects

At the beginning of each chapter in this book, a section called You Are the Designer appeared in which you were asked to imagine that you were the designer of some device or system. Choose any of those projects.

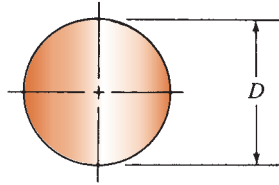
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APPENDIX 17	Hardness Conversion Table
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APPENDIX 1 Properties of Areas

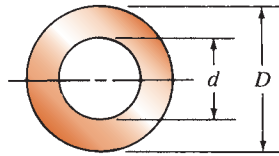
 A = area I = moment of inertia S = section modulus r = radius of gyration = $\sqrt{I/A}$ J = polar moment of inertia Z_p = polar section modulus

(a) Circle



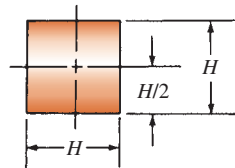
$$\begin{aligned} A &= \pi D^2/4 & r &= D/4 \\ I &= \pi D^4/64 & J &= \pi D^4/32 \\ S &= \pi D^3/32 & Z_p &= \pi D^3/16 \end{aligned}$$

(b) Hollow circle (tube)



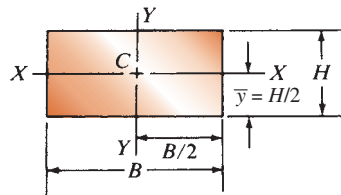
$$\begin{aligned} A &= \pi(D^2 - d^2)/4 & r &= \sqrt{(D^2 + d^2)/4} \\ I &= \pi(D^4 - d^4)/64 & J &= \pi(D^4 - d^4)/32 \\ S &= \pi(D^4 - d^4)/32D & Z_p &= \pi(D^4 - d^4)/16D \end{aligned}$$

(c) Square



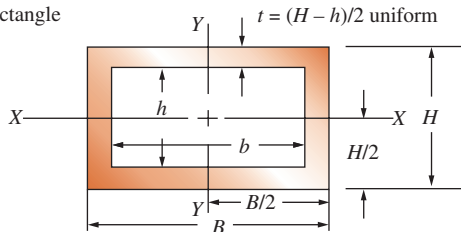
$$\begin{aligned} A &= H^2 & r &= H/\sqrt{12} \\ I &= H^4/12 \\ S &= H^3/6 \end{aligned}$$

(d) Rectangle



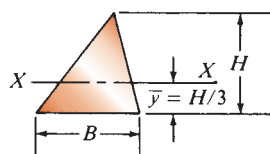
$$\begin{aligned} A &= BH & r_x &= H/\sqrt{12} \\ I_x &= BH^3/12 & r_y &= B/\sqrt{12} \\ I_y &= HB^3/12 \\ S_x &= BH^2/6 \\ S_y &= HB^2/6 \end{aligned}$$

(e) Hollow rectangle



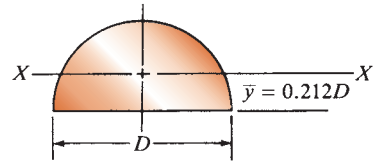
$$\begin{aligned} A &= BH - bh \\ I_x &= \frac{BH^3 - bh^3}{12} & S_x &= \frac{BH^3 - bh^3}{6H} & r_x &= 0.289 \sqrt{\frac{BH^3 - bh^3}{BH - bh}} \\ I_y &= \frac{HB^3 - hb^3}{12} & S_y &= \frac{HB^3 - hb^3}{6B} & r_y &= 0.289 \sqrt{\frac{HB^3 - hb^3}{HB - hb}} \end{aligned}$$

(f) Triangle



$$\begin{aligned} A &= BH/2 & r &= H/\sqrt{18} \\ I &= BH^3/36 \\ S &= BH^2/24 \end{aligned}$$

(g) Semicircle



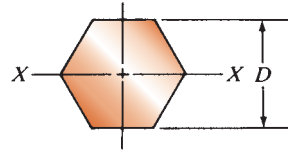
$$A = \pi D^2/8$$

$$I = 0.007D^4$$

$$S = 0.024D^3$$

$$r = 0.132D$$

(h) Regular hexagon



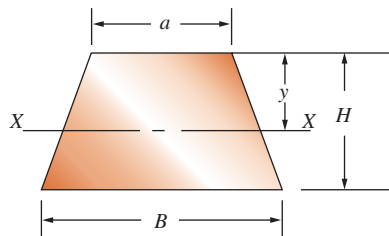
$$A = 0.866D^2$$

$$I = 0.06D^4$$

$$S = 0.12D^3$$

$$r = 0.264D$$

(i) Trapezoid



$$A = H(a + B)/2$$

$$y = \frac{H(a + 2B)}{3(a + B)}$$

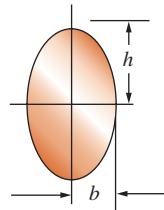
$$I_x = \frac{H^3(a^2 + 4aB + B^2)}{36(a + B)}$$

$$S = \frac{H^2(a^2 + 4aB + B^2)}{12(a + 2B)}$$

$$r = \frac{H^2(a^2 + 4aB + B^2)}{18(a + B)^2}$$

y = Maximum distance from x -axis to outer surface of section

(j) Ellipse



$$A = \pi bh$$

$$I = \frac{\pi h^3 b}{4}$$

$$S = \frac{\pi h^2 b}{4}$$

$$r = h/2$$

APPENDIX 2 Preferred Basic Sizes and Screw Threads

TABLE A2-1 Preferred Basic Sizes

Fractional (in)				Decimal (in)			SI metric (mm)	
1/64	0.015 625	5	5.000	0.010	2.00	8.50	1.0	40
1/32	0.031 25	5 $\frac{1}{4}$	5.250	0.012	2.20	9.00	1.1	45
1/16	0.0625	5 $\frac{1}{2}$	5.500	0.016	2.40	9.50	1.2	50
3/32	0.093 75	5 $\frac{3}{4}$	5.750	0.020	2.60	10.00	1.4	55
1/8	0.1250	6	6.000	0.025	2.80	10.50	1.6	60
5/32	0.156 25	6 $\frac{1}{2}$	6.500	0.032	3.00	11.00	1.8	70
3/16	0.1875	7	7.000	0.040	3.20	11.50	2.0	80
1/4	0.2500	7 $\frac{1}{2}$	7.500	0.05	3.40	12.00	2.2	90
5/16	0.3125	8	8.000	0.06	3.60	12.50	2.5	100
3/8	0.3750	8 $\frac{1}{2}$	8.500	0.08	3.80	13.00	2.8	110
7/16	0.4375	9	9.000	0.10	4.00	13.50	3.0	120
1/2	0.5000	9 $\frac{1}{2}$	9.500	0.12	4.20	14.00	3.5	140
9/16	0.5625	10	10.000	0.16	4.40	14.50	4.0	160
5/8	0.6250	10 $\frac{1}{2}$	10.500	0.20	4.60	15.00	4.5	180
11/16	0.6875	11	11.000	0.24	4.80	15.50	5.0	200
3/4	0.7500	11 $\frac{1}{2}$	11.500	0.30	5.00	16.00	5.5	220
7/8	0.8750	12	12.000	0.40	5.20	16.50	6	250
1	1.000	12 $\frac{1}{2}$	12.500	0.50	5.40	17.00	7	280
1 $\frac{1}{4}$	1.250	13	13.000	0.60	5.60	17.50	8	300
1 $\frac{1}{2}$	1.500	13 $\frac{1}{2}$	13.500	0.80	5.80	18.00	9	350
1 $\frac{3}{4}$	1.750	14	14.000	1.00	6.00	18.50	10	400
2	2.000	14 $\frac{1}{2}$	14.500	1.20	6.50	19.00	11	450
2 $\frac{1}{4}$	2.250	15	15.000	1.40	7.00	19.50	12	500
2 $\frac{1}{2}$	2.500	15 $\frac{1}{2}$	15.500	1.60	7.50	20.00	14	550
2 $\frac{3}{4}$	2.750	16	16.000	1.80	8.00		16	600
3	3.000	16 $\frac{1}{2}$	16.500				18	700
3 $\frac{1}{4}$	3.250	17	17.000				20	800
3 $\frac{1}{2}$	3.500	17 $\frac{1}{2}$	17.500				22	900
3 $\frac{3}{4}$	3.750	18	18.000				25	1000
4	4.000	18 $\frac{1}{2}$	18.500				28	
4 $\frac{1}{4}$	4.250	19	19.000				30	
4 $\frac{1}{2}$	4.500	19 $\frac{1}{2}$	19.500				35	
4 $\frac{3}{4}$	4.750	20	20.000					

TABLE A2-2 American Standard Screw Threads

A. American Standard thread dimensions, numbered sizes

Basic major diameter, D (in)	Coarse threads: UNC		Fine threads: UNF	
	Size-Threads per inch, n	Tensile stress area (in ²)	Size-Threads per inch, n	Tensile stress area (in ²)
0.0600	—	—	0-80	0.001 80
0.0730	1-64	0.002 63	1-72	0.002 78
0.0860	2-56	0.003 70	2-64	0.003 94
0.0990	3-48	0.004 87	3-56	0.005 23
0.1120	4-40	0.006 04	4-48	0.006 61
0.1250	5-40	0.007 96	5-44	0.008 30
0.1380	6-32	0.009 09	6-40	0.010 15
0.1640	8-32	0.0140	8-36	0.014 74
0.1900	10-24	0.0175	10-32	0.0200
0.2160	12-24	0.0242	12-28	0.0258

B. American Standard thread dimensions, fractional sizes

Basic major diameter, D (in)	Coarse threads: UNC		Fine threads: UNF	
	Size-Threads per inch, n	Tensile stress area (in ²)	Size-Threads per inch, n	Tensile stress area (in ²)
0.2500	1/4-20	0.0318	1/4-28	0.0364
0.3125	5/16-18	0.0524	5/16-24	0.0580
0.3750	3/8-16	0.0775	3/8-24	0.0878
0.4375	7/16-14	0.1063	7/16-20	0.1187
0.5000	1/2-13	0.1419	1/2-20	0.1599
0.5625	9/16-12	0.182	9/16-18	0.203
0.6250	5/8-11	0.226	5/8-18	0.256
0.7500	3/4-10	0.334	3/4-16	0.373
0.8750	7/8-9	0.462	7/8-14	0.509
1.000	1 $\frac{1}{8}$ -8	0.606	1 $\frac{1}{8}$ -12	0.663
1.125	1 $\frac{1}{8}$ -7	0.763	1 $\frac{1}{8}$ -12	0.856
1.250	1 $\frac{1}{4}$ -7	0.969	1 $\frac{1}{4}$ -12	1.073
1.375	1 $\frac{3}{8}$ -6	1.155	1 $\frac{3}{8}$ -12	1.315
1.500	1 $\frac{1}{2}$ -6	1.405	1 $\frac{1}{2}$ -12	1.581
1.750	1 $\frac{3}{4}$ -5	1.90		
2.000	2-4 $\frac{1}{2}$	2.50		

TABLE A2-3 Metric Sizes of Screw Threads

Basic major diameter, D (mm)	Coarse threads		Fine threads	
	Basic thread designation			
	$MD \times \text{Pitch}$ (mm) (mm)	Tensile stress area (mm ²)	$MD \times \text{Pitch}$ (mm) (mm)	Tensile stress area (mm ²)
1	M1 $\times$ 0.25	0.460	—	—
1.6	M1.6 $\times$ 0.35	1.27	M1.6 $\times$ 0.20	1.57
2	M2 $\times$ 0.4	2.07	M2 $\times$ 0.25	2.45
2.5	M2.5 $\times$ 0.45	3.39	M2.5 $\times$ 0.35	3.70
3	M3 $\times$ 0.5	5.03	M3 $\times$ 0.35	5.61
4	M4 $\times$ 0.7	8.78	M4 $\times$ 0.5	9.79
5	M5 $\times$ 0.8	14.2	M5 $\times$ 0.5	16.1
6	M6 $\times$ 1	20.1	M6 $\times$ 0.75	22.0
8	M8 $\times$ 1.25	36.6	M8 $\times$ 1	39.2
10	M10 $\times$ 1.5	58.0	M10 $\times$ 1.25	61.2
12	M12 $\times$ 1.75	84.3	M12 $\times$ 1.25	92.1
16	M16 $\times$ 2	157	M16 $\times$ 1.5	167
20	M20 $\times$ 2.5	245	M20 $\times$ 1.5	272
24	M24 $\times$ 3	353	M24 $\times$ 2	384
30	M30 $\times$ 3.5	561	M30 $\times$ 2	621
36	M36 $\times$ 4	817	M36 $\times$ 3	865
42	M42 $\times$ 4.5	1121		
48	M48 $\times$ 5	1473		

APPENDIX 3 Design Properties of Carbon and Alloy Steels

Material designation (SAE number)	Condition	Tensile strength		Yield strength		Ductility (percent elongation in 2 in)	Brinell hardness (HB)
		(ksi)	(MPa)	(ksi)	(MPa)		
1020	Hot-rolled	55	379	30	207	25	111
1020	Cold-drawn	61	420	51	352	15	122
1020	Annealed	60	414	43	296	38	121
1040 ¹	Hot-rolled	72	496	42	290	18	144
1040	Cold-drawn	80	552	71	490	12	160
1040	OQT 1300	88	607	61	421	33	183
1040	OQT 400	113	779	87	600	19	262
1050	Hot-rolled	90	620	49	338	15	180
1050	Cold-drawn	100	690	84	579	10	200
1050	OQT 1300	96	662	61	421	30	192
1050	OQT 400	143	986	110	758	10	321
1117	Hot-rolled	65	448	40	276	33	124
1117	Cold-drawn	80	552	65	448	20	138
1117	WQT 350	89	614	50	345	22	178
1137	Hot-rolled	88	607	48	331	15	176
1137	Cold-drawn	98	676	82	565	10	196
1137	OQT 1300	87	600	60	414	28	174
1137	OQT 400	157	1083	136	938	5	352
1144 ¹	Hot-rolled	94	648	51	352	15	188
1144	Cold-drawn	100	690	90	621	10	200
1144	OQT 1300	96	662	68	469	25	200
1144	OQT 400	127	876	91	627	16	277
1213	Hot-rolled	55	379	33	228	25	110
1213	Cold-drawn	75	517	58	340	10	150
12L13	Hot-rolled	57	393	34	234	22	114
12L13	Cold-drawn	70	483	60	414	10	140
1340 ¹	Annealed	102	703	63	434	26	207
1340	OQT 1300	100	690	75	517	25	235
1340	OQT 1000	144	993	132	910	17	363
1340	OQT 700	221	1520	197	1360	10	444
1340	OQT 400	285	1960	234	1610	8	578
3140	Annealed	95	655	67	462	25	187
3140	OQT 1300	115	792	94	648	23	233
3140	OQT 1000	152	1050	133	920	17	311
3140	OQT 700	220	1520	200	1380	13	461
3140	OQT 400	280	1930	248	1710	11	555
4130	Annealed	81	558	52	359	28	156
4130	WQT 1300	98	676	89	614	28	202
4130	WQT 1000	143	986	132	910	16	302
4130	WQT 700	208	1430	180	1240	13	415
4130	WQT 400	234	1610	197	1360	12	461
4140 ¹	Annealed	95	655	54	372	26	197
4140	OQT 1300	117	807	100	690	23	235
4140	OQT 1000	168	1160	152	1050	17	341
4140	OQT 700	231	1590	212	1460	13	461
4140	OQT 400	290	2000	251	1730	11	578
4150	Annealed	106	731	55	379	20	197
4150	OQT 1300	127	880	116	800	20	262
4150	OQT 1000	197	1360	181	1250	11	401
4150	OQT 700	247	1700	229	1580	10	495
4150	OQT 400	300	2070	248	1710	10	578

(Continued)

APPENDIX 3 (Continued)

Material designation (SAE number)	Condition	Tensile strength		Yield strength		Ductility (percent elongation in 2 in)	Brinell hardness (HB)
		(ksi)	(MPa)	(ksi)	(MPa)		
4340 ¹	Annealed	108	745	68	469	22	217
4340	OQT 1300	140	965	120	827	23	280
4340	OQT 1000	171	1180	158	1090	16	363
4340	OQT 700	230	1590	206	1420	12	461
4340	OQT 400	283	1950	228	1570	11	555
5140	Annealed	83	572	42	290	29	167
5140	OQT 1300	104	717	83	572	27	207
5140	OQT 1000	145	1000	130	896	18	302
5140	OQT 700	220	1520	200	1380	11	429
5140	OQT 400	276	1900	226	1560	7	534
5150	Annealed	98	676	52	359	22	197
5150	OQT 1300	116	800	102	700	22	241
5150	OQT 1000	160	1100	149	1030	15	321
5150	OQT 700	240	1650	220	1520	10	461
5150	OQT 400	312	2150	250	1720	8	601
5160	Annealed	105	724	40	276	17	197
5160	OQT 1300	115	793	100	690	23	229
5160	OQT 1000	170	1170	151	1040	14	341
5160	OQT 700	263	1810	237	1630	9	514
5160	OQT 400	322	2220	260	1790	4	627
6150 ¹	Annealed	96	662	59	407	23	197
6150	OQT 1300	118	814	107	738	21	241
6150	OQT 1000	183	1260	173	1190	12	375
6150	OQT 700	247	1700	223	1540	10	495
6150	OQT 400	315	2170	270	1860	7	601
8650	Annealed	104	717	56	386	22	212
8650	OQT 1300	122	841	113	779	21	255
8650	OQT 1000	176	1210	155	1070	14	363
8650	OQT 700	240	1650	222	1530	12	495
8650	OQT 400	282	1940	250	1720	11	555
8740	Annealed	100	690	60	414	22	201
8740	OQT 1300	119	820	100	690	25	241
8740	OQT 1000	175	1210	167	1150	15	363
8740	OQT 700	228	1570	212	1460	12	461
8740	OQT 400	290	2000	240	1650	10	578
9255	Annealed	113	780	71	490	22	229
9255	Q&T 1300	130	896	102	703	21	262
9255	Q&T 1000	181	1250	160	1100	14	352
9255	Q&T 700	260	1790	240	1650	5	534
9255	Q&T 400	310	2140	287	1980	2	601

Notes: Properties common to all carbon and alloy steels:

Poisson's ratio: 0.27.

Shear modulus: 11.5×10^6 psi; 80 GPa.

Coefficient of thermal expansion: 6.5×10^{-6} F⁻¹.

Density: 0.283 lb/in³; 7680 kg/m³.

Modulus of elasticity: 30×10^6 psi; 207 GPa.

¹See Appendix 4 for graphs of properties versus heat treatment.

APPENDIX 4 Properties of Heat-Treated Steels

Treatment: Normalized at 1650°F; reheated to 1550°F; quenched in water.
1-in Round Treated; 0.505-in Round Tested. As-quenched HB 534.

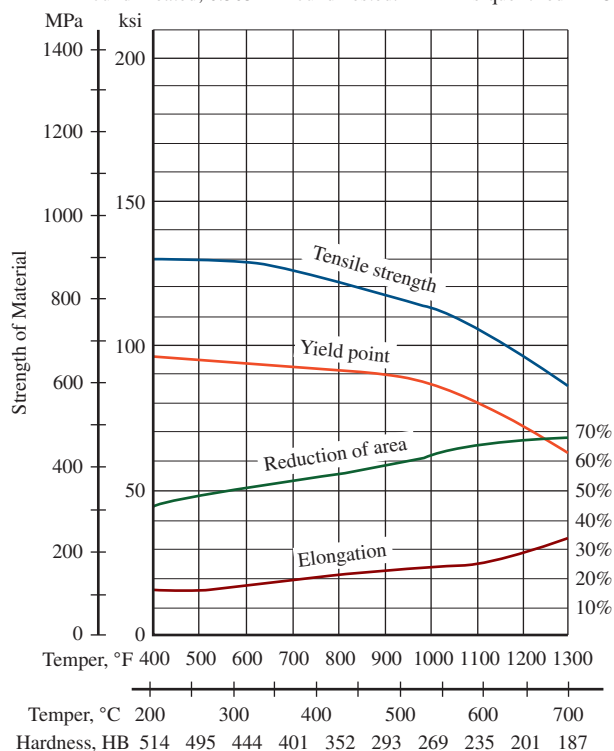


FIGURE A4-1 Properties of heat-treated SAE 1040, water-quenched and tempered

Treatment: Normalized at 1650°F; reheated to 1550°F; quenched in oil.
1-in Round Treated; 0.505-in Round Tested. As-quenched HB 285.

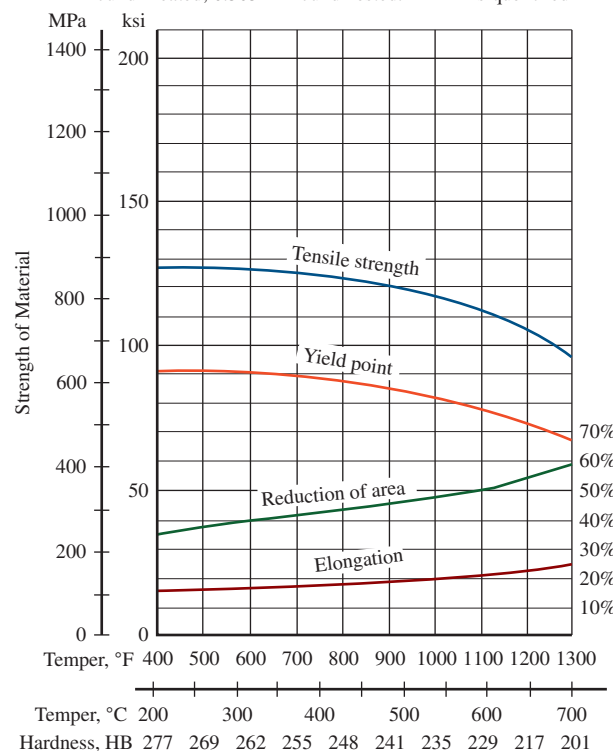


FIGURE A4-2 Properties of heat-treated SAE 1144, oil-quenched and tempered

Treatment: Normalized at 1600°F; reheated to 1525°F; quenched in agitated oil.
0.565-in Round Treated; 0.505-in Round Tested. As-quenched HB 601.

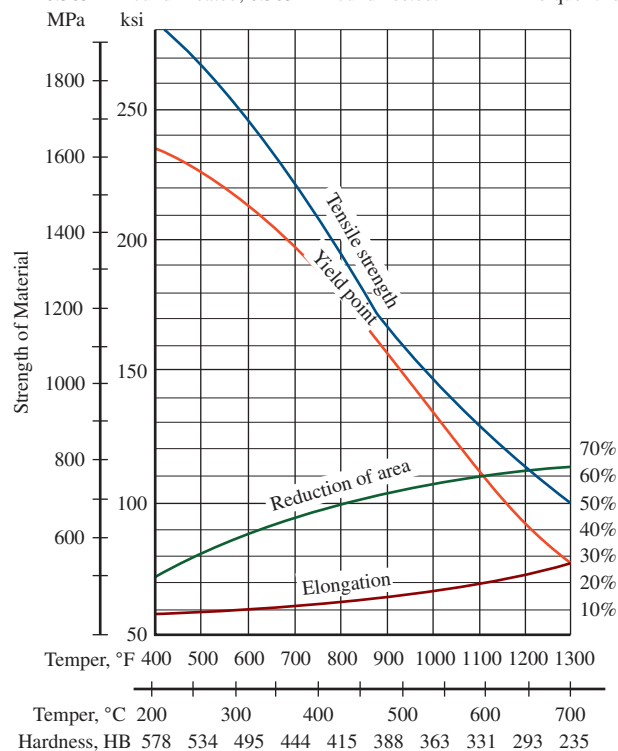


FIGURE A4-3 Properties of heat-treated SAE 1340, oil-quenched and tempered

Treatment: Normalized at 1600°F; reheated to 1550°F; quenched in agitated oil.
0.530-in Round Treated; 0.505-in Round Tested. As-quenched HB 601.

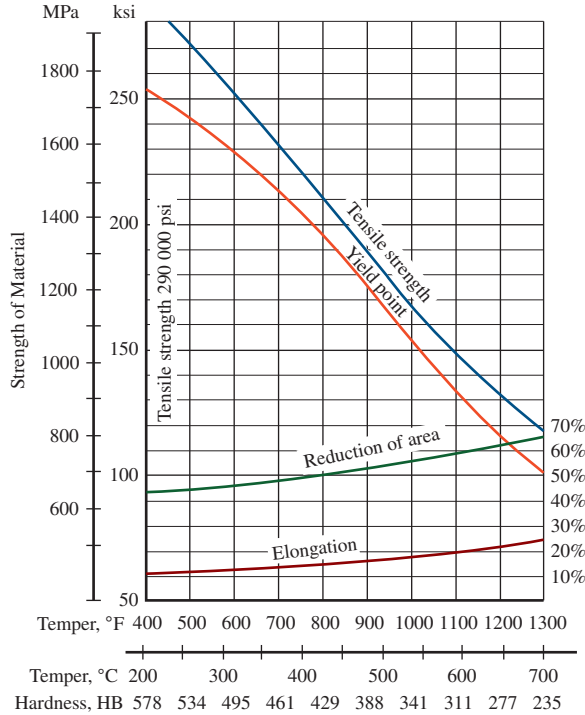


FIGURE A4-4 Properties of heat-treated SAE 4140, oil-quenched and tempered

Treatment: Normalized at 1600°F; reheated to 1475°F; quenched in agitated oil.
0.530-in Round Treated; 0.505-in Round Tested. As-quenched HB 601.

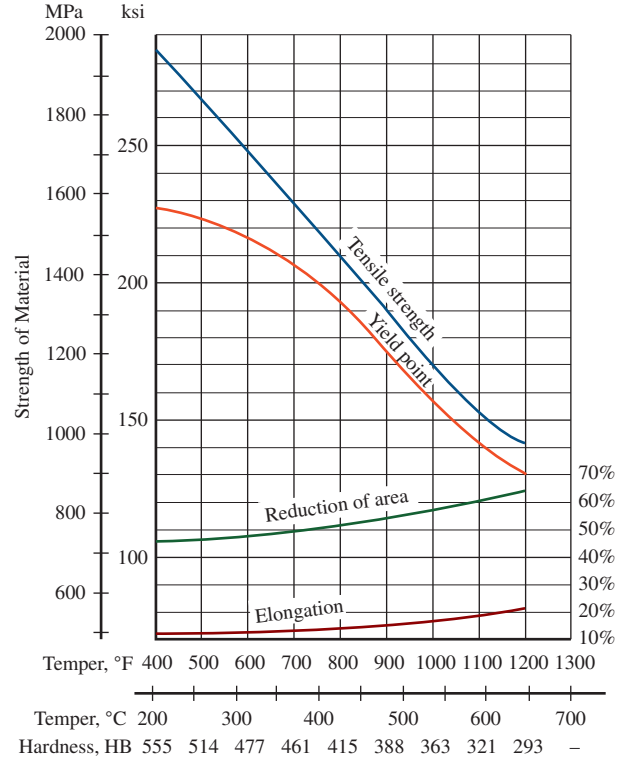


FIGURE A4-5 Properties of heat-treated SAE 4340, oil-quenched and tempered

Treatment: Normalized at 1600°F; reheated to 1550°F; quenched in agitated oil.
0.565-in Round Treated; 0.505-in Round Tested. As-quenched HB 627.

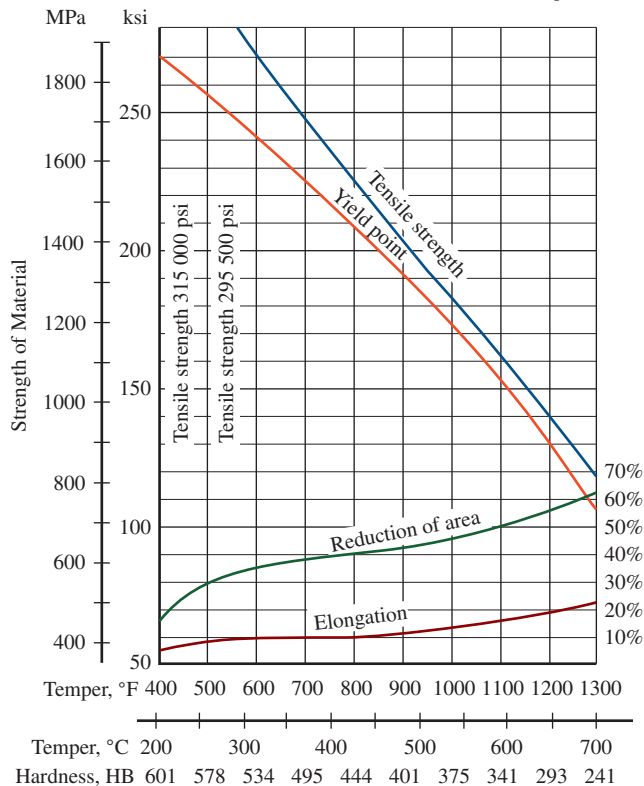


FIGURE A4-6 Properties of heat-treated SAE 6150, oil-quenched and tempered

APPENDIX 5 Properties of Carburized Steels

Material designation (SAE number)	Condition	Core properties						
		Tensile strength		Yield strength		Ductility (percent elongation in 2 in)	Brinell hardness (HB)	Case hardness (HRC)
		(ksi)	(MPa)	(ksi)	(MPa)			
1015	SWQT 350	106	731	60	414	15	217	62
1020	SWQT 350	129	889	72	496	11	255	62
1022	SWQT 350	135	931	75	517	14	262	62
1117	SWQT 350	125	862	66	455	10	235	65
1118	SWQT 350	144	993	90	621	13	285	61
4118	SOQT 300	143	986	93	641	17	293	62
4118	DOQT 300	126	869	63	434	21	241	62
4118	SOQT 450	138	952	89	614	17	277	56
4118	DOQT 450	120	827	63	434	22	229	56
4320	SOQT 300	218	1500	178	1230	13	429	62
4320	DOQT 300	151	1040	97	669	19	302	62
4320	SOQT 450	211	1450	173	1190	12	415	59
4320	DOQT 450	145	1000	94	648	21	293	59
4620	SOQT 300	119	820	83	572	19	277	62
4620	DOQT 300	122	841	77	531	22	248	62
4620	SOQT 450	115	793	80	552	20	248	59
4620	DOQT 450	115	793	77	531	22	235	59
4820	SOQT 300	207	1430	167	1150	13	415	61
4820	DOQT 300	204	1405	165	1140	13	415	60
4820	SOQT 450	205	1410	184	1270	13	415	57
4820	DOQT 450	196	1350	171	1180	13	401	56
8620	SOQT 300	188	1300	149	1030	11	388	64
8620	DOQT 300	133	917	83	572	20	269	64
8620	SOQT 450	167	1150	120	827	14	341	61
8620	DOQT 450	130	896	77	531	22	262	61
E9310	SOQT 300	173	1190	135	931	15	363	62
E9310	DOQT 300	174	1200	139	958	15	363	60
E9310	SOQT 450	168	1160	137	945	15	341	59
E9310	DOQT 450	169	1170	138	952	15	352	58

Notes: Properties given are for a single set of tests on 1/2-in round bars.

SWQT: single water-quenched and tempered.

SOQT: single oil-quenched and tempered.

DOQT: double oil-quenched and tempered.

300 and 450 are the tempering temperatures in °F. Steel was carburized for 8 h. Case depth ranged from 0.045 to 0.075 in.

APPENDIX 6 Properties of Stainless Steels

Material designation		Condition	Tensile strength		Yield strength		Ductility (percent elongation in 2 in)
SAE number	UNS		(ksi)	(MPa)	(ksi)	(MPa)	
Austenitic steels							
201	S20100	Annealed	115	793	55	379	55
		1/4 hard	125	862	75	517	20
		1/2 hard	150	1030	110	758	10
		3/4 hard	175	1210	135	931	5
		Full hard	185	1280	140	966	4
301	S30100	Annealed	110	758	40	276	60
		1/4 hard	125	862	75	517	25
		1/2 hard	150	1030	110	758	15
		3/4 hard	175	1210	135	931	12
		Full hard	185	1280	140	966	8
304	S30400	Annealed	85	586	35	241	60
310	S31000	Annealed	95	655	45	310	45
316	S31600	Annealed	80	552	30	207	60
Ferritic steels							
405	S40500	Annealed	70	483	40	276	30
430	S43000	Annealed	75	517	40	276	30
		Full hard	90	621	80	552	15
446	S44600	Annealed	80	552	50	345	25
Martensitic steels							
410	S41000	Annealed	75	517	40	276	30
416	S41600	Q&T 600	180	1240	140	966	15
		Q&T 1000	145	1000	115	793	20
		Q&T 1400	90	621	60	414	30
		Q&T 1700	75	517	50	345	40
431	S43100	Q&T 600	195	1344	150	1034	15
440A	S44002	Q&T 600	280	1930	270	1860	3
501	S50100	Annealed	70	483	30	207	28
		OQT 1000	175	1210	135	931	15
Precipitation-hardening steels							
17-4PH	S17400	H 900	210	1450	185	1280	14
		H 1150	145	1000	125	862	19
17-7PH	S17700	RH 950	200	1380	175	1210	10
		TH 1050	175	1210	155	1070	12
PH 13-8 Mo	S13800	H 950	220	1517	205	1413	10
		H 1050	175	1207	165	1138	12
		H 1150	135	931	90	621	14

APPENDIX 7 Properties of Structural Steels

Material ASTM no. and products	Ultimate strength, s_u ¹		Yield strength, s_y ¹		Percent elongation in 2 in
	ksi	MPa	ksi	MPa	
A36—Carbon steel: shapes, plates, and bars	58	400	36	248	21
A 53—grade B-pipe	60	414	35	240	23
A242—HSLA corrosion resistant: shapes, plates, and bars					
≤ $\frac{3}{4}$ in thick	70	483	50	345	21
$\frac{3}{4}$ to $1\frac{1}{2}$ in thick	67	462	46	317	21
$1\frac{1}{2}$ to 4 in thick	63	434	42	290	21
A500—Cold-formed structural tubing					
Round, grade B	58	400	42	290	23
Round, grade C	62	427	46	317	21
Shaped, grade B	58	400	46	317	23
Shaped, grade C	65	427	50	345	21
A501—Hot-formed structural tubing, round or shaped	58	400	36	248	23
A514—Quenched and tempered alloy steel: plate					
≤ $2\frac{1}{2}$ in thick	110	758	100	690	18
$2\frac{1}{2}$ to 6 in thick	100	690	90	620	16
A572—HSLA columbium-vanadium steel: shapes, plates, and bars					
Grade 42	60	414	42	290	24
Grade 50	65	448	50	345	21
Grade 60	75	517	60	414	18
Grade 65	80	552	65	448	17
A913—HSLA, grade 65: shapes	80	552	65	448	17
A992—HSLA: W-Shapes only	65	448	50	345	21

Notes: ¹Minimum values; may range higher.

HSLA—High strength low-alloy.

The American Institute of Steel Construction specifies $E = 29 \times 10^6$ psi (200 GPa) for structural steel.

APPENDIX 8 Design Properties of Cast Iron—U.S. Units Basis³

U.S. units	Ultimate strength						Yield strength		¹ Modulus of elasticity, <i>E</i>		Percent elongation
Material type and grade	<i>s_u</i>		¹ <i>s_{uc}</i>		¹ <i>s_{us}</i>		<i>s_{yt}</i>		¹ Modulus of elasticity, <i>E</i>		Percent elongation
	ksi	² MPa	ksi	² MPa	ksi	² MPa	ksi	² MPa	10 ⁶ psi	² GPa	%
Gray iron—ASTM A48											
No. 20A	20	138	80	552	32	221	—	—	12.2	84	<1.0
No. 30A	30	207	113	779	47	324	—	—	16.9	117	<1.0
No. 40A	40	276	140	965	57	393	—	—	19.4	134	<1.0
No. 50A	50	345	158	1089	66	455	—	—	20.8	143	<1.0
No. 60A	60	414	170	1172	72	496	—	—	21.5	148	<1.0
Ductile iron—ASTM A536											
60-40-18	60	414	—	—	57	393	40	276	24	165	18
65-45-12	65	448	—	—	—	—	45	310	24	165	12
80-55-08	80	552	—	—	—	—	55	379	24	165	6
100-70-03	100	690	—	—	—	—	70	483	24	165	3
120-90-02	120	827	180	1241	—	—	90	621	23	159	2
Austempered ductile iron (ADI)—ASTM A897											
110/70/11	110	758	—	—	—	—	70	483	22	152	11
130/90/09	130	896	—	—	—	—	90	621	22	152	9
150/110/07	150	1034	—	—	—	—	110	758	22	152	7
175/125/04	175	1207	—	—	—	—	125	862	22	152	4
200/155/02	200	1379	—	—	—	—	155	1069	22	152	2
230/185/01	230	1586	—	—	—	—	185	1276	22	152	1
Ferritic malleable iron—ASTM A47											
32510	50	345	—	—	—	—	32.5	224	25	172	10
Pearlitic malleable iron—ASTM A220											
40010	60	414	240	1655	43	296	40	276	26	179	10
45008	65	448	240	1655	49	338	45	310	26	179	8
45006	65	448	240	1655	49	338	45	310	26	179	6
50005	70	483	240	1655	55	379	50	345	26	179	5
60004	80	552	240	1655	65	448	60	414	26	179	4
70003	85	586	240	1655	68	469	70	483	26	179	3
80002	95	655	240	1655	75	517	80	552	26	179	2
90001	105	724	240	1655	78	538	90	621	26	179	1

Notes: ¹Approximate values; not part of the standards; If critical, negotiate with supplier.

²Metric data computed using: (6.895 × U.S. data); Not part of the standards.

³The density of cast irons ranges from 0.25 lb_m/in³ to 0.27 lb_m/in³ (6920 kg/m³ to 7480 kg/m³).

APPENDIX 8A Design Properties of Cast Iron—SI Units Basis³

SI units	Ultimate strength						Yield strength		¹ Modulus of elasticity, <i>E</i>		Percent elongation
Material type and grade	<i>s_u</i>		¹ <i>s_{uc}</i>		¹ <i>s_{us}</i>		<i>s_{yt}</i>		¹ Modulus of elasticity, <i>E</i>		Percent elongation
	² ksi	MPa	² ksi	MPa	² ksi	MPa	² ksi	MPa	10 ⁶ psi	GPa	%
Gray iron—ASTM A48M											
No. 150A	22	150	—	—	—	—	—	—	12.2	84	<1.0
No. 200A	29	200	—	—	—	—	—	—	16.9	117	<1.0
No. 275A	40	275	—	—	—	—	—	—	19.4	134	<1.0
No. 350A	51	350	—	—	—	—	—	—	20.8	143	<1.0
No. 400A	58	400	—	—	—	—	—	—	21.5	148	<1.0
Ductile iron—ASTM A536 <i>Note: No metric grades included in the standard.</i>											
Austempered ductile iron (ADI)—ASTM A897M											
750/500/11	109	750	—	—	—	—	73	500	22	152	11
900/650/09	131	900	—	—	—	—	94	650	22	152	9
1050/750/07	152	1050	—	—	—	—	109	750	22	152	7
1200/850/04	174	1200	—	—	—	—	123	850	22	152	4
1400/1100/02	203	1400	—	—	—	—	160	1100	22	152	2
1600/1300/01	232	1600	—	—	—	—	189	1300	22	152	1
Ferritic malleable iron—ASTM A47M											
22010	49	340	—	—	—	—	32	220	25	172	10
Pearlitic malleable iron—ASTM A220M											
280M10	58	400	—	—	—	—	41	280	26	179	10
310M8	65	450	—	—	—	—	45	310	26	179	8
310M6	65	450	—	—	—	—	45	310	26	179	6
340M5	70	480	—	—	—	—	49	340	26	179	5
410M4	80	550	—	—	—	—	59	410	26	179	4
480M3	86	590	—	—	—	—	70	480	26	179	3
550M2	94	650	—	—	—	—	80	550	26	179	2
620M1	104	720	—	—	—	—	90	620	26	179	1

Notes: Five additional intermediate grades are included in standard A48M for gray iron.

¹Approximate values; not part of the standards; If critical, negotiate with supplier.

²U.S. data computed using: (SI data/6.895); Not part of the standards.

³The density of cast irons ranges from 6920 kg/m³ to 7480 kg/m³ (0.25 lb_m/in³ to 0.27 lb_m/in³).

APPENDIX 9 Typical Properties of Aluminum

Alloy and temper	Tensile strength		Yield strength		Ductility (percent elongation in 2 in)	Shearing strength		Endurance strength	
	(ksi)	(MPa)	(ksi)	(MPa)		(ksi)	(MPa)	(ksi)	(MPa)
1060-O	10	69	4	28	43	7	48	3	21
1060-H14	14	97	13	90	12	9	62	5	34
1060-H18	19	131	18	124	6	11	121	6.5	41
1350-O	12	83	4	28	28	8	55		
1350-H14	16	110	14	97		10	69		
1350-H19	27	186	24	165		15	103	7	48
2014-O	27	186	14	97	18	18	124	13	90
2014-T4	62	427	42	290	20	38	262	20	138
2014-T6	70	483	60	414	13	42	290	18	124
2024-O	27	186	11	76	22	18	124	13	90
2024-T4	68	469	47	324	19	41	283	20	138
2024-T361	72	496	57	393	13	42	290	18	124
2219-O	25	172	11	76	18				
2219-T62	60	414	42	290	10			15	103
2219-T87	69	476	57	393	10			15	103
3003-O	16	110	6	41	40	11	121	7	48
3003-H14	22	152	21	145	16	14	97	9	62
3003-H18	29	200	27	186	10	16	110	10	69
5052-O	28	193	13	90	30	18	124	16	110
5052-H34	38	262	31	214	14	21	145	18	124
5052-H38	42	290	37	255	8	24	165	20	138
6061-O	18	124	8	55	30	12	83	9	62
6061-T4	35	241	21	145	25	24	165	14	97
6061-T6	45	310	40	276	17	30	207	14	97
6063-O	13	90	7	48		10	69	8	55
6063-T4	25	172	13	90	22				
6063-T6	35	241	31	214	12	22	152	10	69
7178-O	33	228	15	103	16				
7178-T6	88	607	78	538	11				
7075-O	33	228	15	103	16	22	152		
7075-T6	83	572	73	503	11	48	331	23	159
Casting alloys (permanent mold casting)									
204.0-14	48	331	29	200	8	—	—		
206.0-T6	65	445	59	405	6	—	—		
356.0-T6	41	283	30	207	10	—	—		

Note: Common properties:
Density: 0.095 to 0.102 lb/in³ (2635 to 2829 kg/m³).
Endurance strength at 5 × 10⁸ cycles.

Modulus of elasticity	Alloys
10.0×10 ⁶ psi (69.0 GPa)	1100, 3003, 6061, 6063
10.2×10 ⁶ psi (70.3 GPa)	5154
10.4×10 ⁶ psi (71.7 GPa)	7075
10.6×10 ⁶ psi (73.1 GPa)	2014

APPENDIX 10-1 Properties of Die-Cast Zinc Alloys

Material designation	Tensile strength		Yield strength		Compression strength		Percent elongation	Density		Modulus of elasticity		Impact strength	
	(ksi)	(MPa)	(ksi)	(MPa)	(ksi)	(MPa)		(%)	(lb _m /in ³)	(kg/m ³)	(10 ⁶ psi)	(GPa)	(ft · lb)
ZAMAK #3	41	283	32	221	60	414	10	0.240	6600	12.4	85.5	43.0	58
ZAMAK #5	48	331	33	228	87	600	7	0.240	6600	12.4	85.5	48.0	65
ZA-8	54	374	42	290	37	252	8	0.227	6300	12.4	85.5	31.0	42
ZA-12	59	404	46	320	39	269	5.5	0.216	6000	12.0	82.7	21.0	29
ZA-27	61	421	55	379	52	385	2	0.181	5000	11.3	77.9	9.0	5

Note: Strength values are typical average values; may range higher or lower.

APPENDIX 10-2 Properties of Die-Cast Magnesium Alloys

Material designation	Tensile strength		¹ Yield strength		Compression strength		Percent elongation	Density		Modulus of elasticity		Impact strength	
	(ksi)	(MPa)	(ksi)	(MPa)	(ksi)	(MPa)	(%)	(lb _m /in ³)	(kg/m ³)	(10 ⁶ psi)	(GPa)	(ft · lb)	(J or N · m)
AZ-91 MgAl9Zn1(A)	33.4	230	22.5	155	21.5	148	3	0.0654	1810	6.5	45	4.4	6
AM-60 MgAl6Mn	31.9	220	19.6	135	—	—	9	0.0650	1800	6.5	45	12.5	17
AM-50 MgAl5Mn	29.7	205	17.4	120	16.4	113	10	0.0639	1770	6.5	45	13.3	18
AM-20 MgAl2Mn	26.8	185	13.1	90	10.7	74	13	0.0632	1750	6.5	45	13.3	18

Notes: Strength values are typical average values; may range higher or lower.

¹Also called 0.2% proof stress.

APPENDIX 11-1 Properties of Nickel-Based Alloys

Material designation	Tensile strength		Yield strength		Percent elongation	Density		Modulus of elasticity	
	(ksi)	(Mpa)	(ksi)	(MPa)		(lb _m /in ³)	(kg/m ³)	(10 ⁶ psi)	GPa
N06600 Annealed	93	640	37	255	45	0.304	8420	30	207
N06110 40% cold worked	175	1205	150	1034	18	0.302	8330	30	207
N04400 Annealed	80	550	30	207	50	0.318	8800	26	181
N04400 cold drawn	100	690	75	517	30	0.318	8800	26	181

APPENDIX 11-2 Properties of Titanium Alloys

Material designation	Tensile strength		Yield strength		Percent elongation	Density		Modulus of elasticity	
	(ksi)	(MPa)	(ksi)	(MPa)		(lb _m /in ³)	(kg/m ³)	(10 ⁶ psi)	(GPa)
Commercially Pure alpha titanium									
Ti-35A Wrought	35	241	25	172	24	0.163	4515	15.0	103
Ti-50A Wrought	50	345	40	276	20	0.163	4515	15.0	103
Ti-65 Wrought	65	448	55	379	18	0.163	4515	15.0	103
Alpha alloy									
Ti-0.2Pd Wrought	50	345	40	276	20	0.163	4515	14.9	103
Beta alloy									
Ti-3Al-13V-11Cr Air cooled from 1400°F	135	931	130	896	16	0.176	4875	14.7	101
Ti-3Al-13V-11Cr Air cooled from 1400°F and aged	185	1280	175	1210	6	0.176	4875	16.0	110
Alpha-beta alloy									
Ti-6Al-4V Annealed	130	896	120	827	10	0.160	4432	16.5	114
Ti-6Al-4V Quenched and aged at 1000°F	160	1100	150	1030	7	0.160	4432	16.5	114

APPENDIX 12 Properties of Bronzes, Brasses, and Other Copper Alloys

Material	UNS number designation	Tensile strength		Yield strength		Ductility (% elongation)	Modulus of elasticity		Density	
		(ksi)	(MPa)	(ksi)	(MPa)		(10 ⁶ psi)	(GPa)	lb _m /in ³	kg/m ³
Bronzes—Wrought bars, rods										
Leaded phosphor bronze	C54400-H04	68	469	57	393	20	15.0	103	0.320	8890
Silicon bronze	C65500-H06	108	745	60	414	13	15.0	103	0.308	8530
Manganese bronze	C67500-H02	84	579	60	414	19	15.0	103	0.302	8360
Bronzes—Sand cast										
Manganese bronze	C86200-M01	95	655	48	331	20	15.0	103	0.288	7970
Bearing bronze	C93200-M01	35	241	18	124	20	14.5	100	0.322	8910
Aluminum bronze	C95400-M01	85	586	35	241	18	15.5	107	0.269	7450
Copper-nickel-iron alloy	C96200-M01	45	310	25	172	20	18.0	124	0.323	8940
Copper-nickel-zinc alloy (also called nickel silver)	C97300-M01	35	241	17	117	20	16.0	110	0.321	8890
Brasses—Wrought bars, rods										
Extra high-leaded brass	C35600-H02	55	379	25	172	10	97	14.0	0.307	8500
Free-cutting brass	C36000-H04	65	448	30	207	6	103	15.0	0.307	8500
Free-cutting Muntz metal	C37000-H04	80	552	60	414	6	103	15.0	0.304	8410
Naval brass	C46400-H02	75	517	53	365	20	103	15.0	0.304	8410

Notes:

1. Strength and ductility values listed are typical at the approximate middle of available ranges and not guaranteed.
2. Properties vary widely for different section sizes, strain hardening, and thermal treatment conditions.
3. Strain hardening conditions: 1/8 hard (H00), 1/4 hard (H01), 1/2 hard (H02), 3/4 hard (H03), full hard (H04), extra hard (H06), spring (H08), and higher.
4. More details and data for numerous additional alloys can be found at www.copper.org/resources/properties.

APPENDIX 13 Typical Properties of Selected Plastics

Material	Type	Tensile strength		Tensile modulus		Flexural strength		Flexural modulus		Impact strength IZOD (ft · lb/in of notch)
		(ksi)	(MPa)	(ksi)	(MPa)	(ksi)	(MPa)	(ksi)	(MPa)	
¹ Nylon 66 30% Glass	Dry	21.0	146	1200	8700	32.0	221	1100	7900	
	50% R.H.	15.0	102	800	5500					
² ABS	Medium-impact	6.0	41	360	2480	11.5	79	310	2140	4.0
	High-impact	5.0	34	250	1720	8.0	55	260	1790	7.0
Polycarbonate Acrylic	General-purpose	9.0	62	340	2340	11.0	76	300	2070	12.0
	Standard	10.5	72	430	2960	16.0	110	460	3170	0.4
	High-impact	5.4	37	220	1520	7.0	48	230	1590	1.2
³ PVC	Rigid	6.0	41	350	2410			300	2070	0.4–20.0 (varies widely)
Polyimide	25% graphite powder filler	5.7	39			12.8	88	900	6210	0.25
	Glass-fiber filler	27.0	186			50.0	345	3250	22 400	17.0
	Laminate	50.0	345			70.0	483	4000	27 580	13.0
Acetal	Copolymer	8.0	55	410	2830	13.0	90	375	2590	1.3
Polyurethane	Elastomer	5.0	34	100	690	0.6	4			No break
Phenolic	General	6.5	45	1100	7580	9.0	62	1100	7580	0.3
⁴ Polyester with glass-fiber mat reinforcement (approx. 30% glass by weight)										
	Lay-up, contact mold	9.0	62			16.0	110	800	5520	
	Cold-press molded	12.0	83			22.0	152	1300	8960	
	Compression molded	25.0	172			10.0	69	1300	8960	

Notes: ¹Also known as Polyamide 66 or PA 66.

²Acrylonitrile-butadiene-styrene.

³Polyvinyl chloride.

⁴Polyethylene terephthalate (PET) thermoplastic polyester resin.

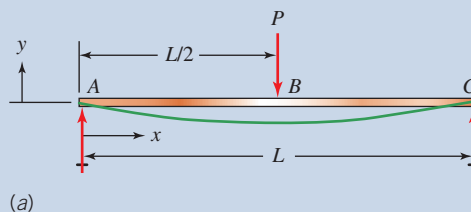
APPENDIX 14 Beam-Deflection Formulas

TABLE A14-1 Beam-Deflection Formulas for Simply Supported Beams

$$y_B = y_{\max} = \frac{-PL^3}{48EI} \text{ at center}$$

Between A and B:

$$y = \frac{-Px}{48EI}(3L^2 - 4x^2)$$



$$y_{\max} = \frac{-Pab(L+b)\sqrt{3a(L+b)}}{27EIL}$$

$$\text{at } x_1 = \sqrt{a(L+b)/3}$$

$$y_B = \frac{-Pa^2b^2}{3EIL} \text{ at load}$$

Between A and B (the longer segment):

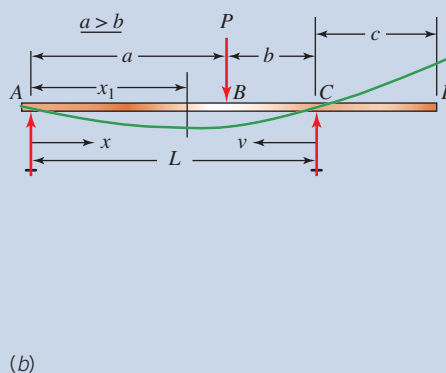
$$y = \frac{-Pbx}{6EIL}(L^2 - b^2 - x^2)$$

Between B and C (the shorter segment):

$$y = \frac{-Pav}{6EIL}(L^2 - v^2 - a^2)$$

At end of overhang at D:

$$y_D = \frac{Pabc}{6EIL}(L+a)$$



$$y_E = y_{\max} = \frac{-Pa}{24EI}(3L^2 - 4a^2) \text{ at center}$$

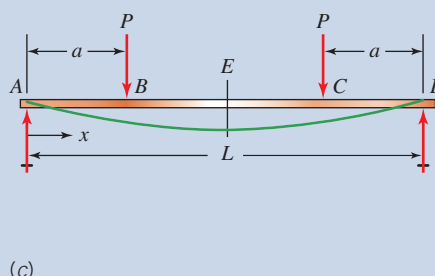
$$y_B = y_C = \frac{-Pa^2}{6EI}(3L - 4a) \text{ at loads}$$

Between A and B:

$$y = \frac{-Px}{6EI}(3aL - 3a^2 - x^2)$$

Between B and C:

$$y = \frac{-Pa}{6EI}(3Lx - 3x^2 - a^2)$$



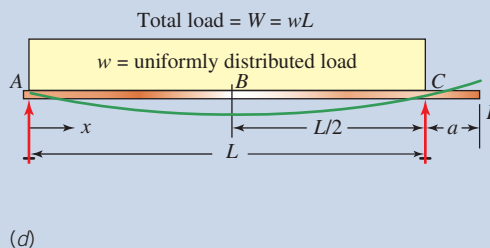
$$y_B = y_{\max} = \frac{-5wL^4}{384EI} = \frac{-5WL^3}{384EI} \text{ at center}$$

Between A and B:

$$y = \frac{-wx}{24EI}(L^3 - 2Lx^2 + x^3)$$

At D at end:

$$y_D = \frac{wL^3a}{24EI}$$



(Continued)

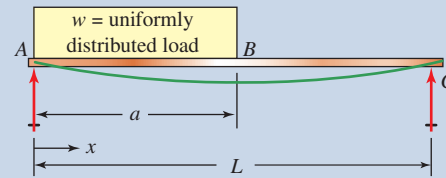
TABLE A14-1 (Continued)

Between A and B:

$$y = \frac{-wx}{24EI} [a^2(2L - a)^2 - 2ax^2(2L - a) + Lx^3]$$

Between B and C:

$$y = \frac{-wa^2(L - x)}{24EI} (4Lx - 2x^2 - a^2)$$



(e)

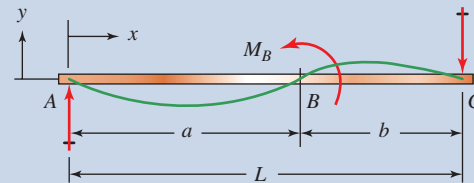
 M_B = concentrated moment at B

Between A and B:

$$y = \frac{-M_B}{6EI} \left[\left(6a - \frac{3a^2}{L} - 2L \right) x - \frac{x^3}{L} \right]$$

Between B and C:

$$y = \frac{M_B}{6EI} \left[3a^2 + 3x^2 - \frac{x^3}{L} - \left(2L + \frac{3a^2}{L} \right) x \right]$$



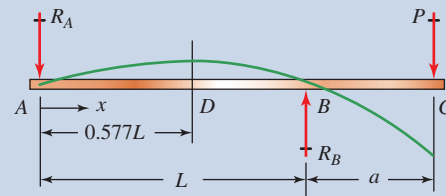
(f)

At C at end of overhang:

$$y_C = \frac{-Pa^2}{3EI} (L + a)$$

At D, maximum upward deflection:

$$y_D = 0.06415 \frac{PaL^2}{EI}$$



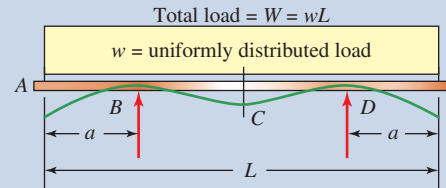
(g)

At C at center:

$$y = \frac{-W(L - 2a)^3}{384EI} \left[\frac{5}{L}(L - 2a) - \frac{24}{L} \left(\frac{a^2}{L - 2a} \right) \right]$$

At A and E at ends:

$$y = \frac{-W(L - 2a)^3 a}{24EI} \left[-1 + 6 \left(\frac{a}{L - 2a} \right)^2 + 3 \left(\frac{a}{L - 2a} \right)^3 \right]$$



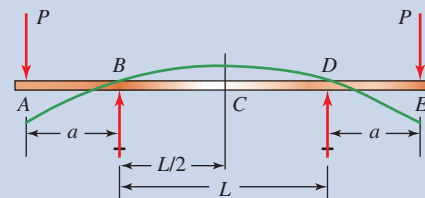
(h)

At C at center:

$$y = \frac{PL^2 a}{8EI}$$

At A and E at ends at loads:

$$y = \frac{-Pa^2}{3EI} \left(a + \frac{3}{2} L \right)$$



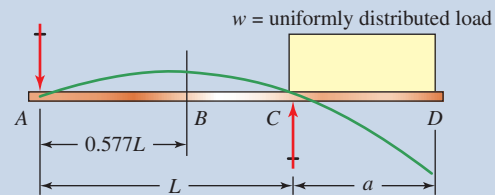
(i)

At B:

$$y = 0.03208 \frac{wa^2 L^2}{EI}$$

At D at end:

$$y = \frac{-wa^3}{24EI} (4L + 3a)$$



(j)

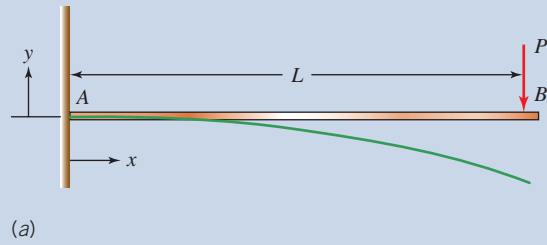
TABLE A14-2 Beam-Deflection Formulas for Cantilevers

At B at end:

$$y_B = y_{\max} = \frac{-PL^3}{3EI}$$

Between A and B:

$$y = \frac{-Px^2}{6EI} (3L - x)$$



At B at load:

$$y_B = \frac{-Pa^3}{3EI}$$

At C at end:

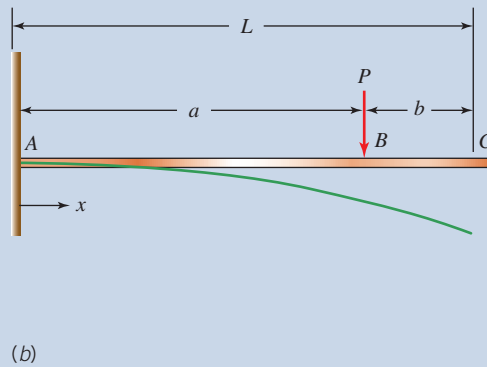
$$y_C = y_{\max} = \frac{-Pa^2}{6EI} (3L - a)$$

Between A and B:

$$y = \frac{-Px^2}{6EI} (3a - x)$$

Between B and C:

$$y = \frac{-Pa^2}{6EI} (3x - a)$$

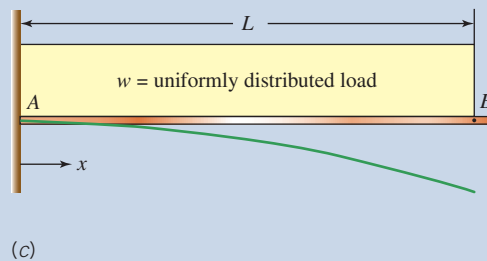

 $W = \text{total load} = wL$

At B at end:

$$y_B = y_{\max} = \frac{-WL^3}{8EI}$$

Between A and B:

$$y = \frac{-Wx^2}{24EIL} [2L^2 + (2L - x)^2]$$


 $M_B = \text{concentrated moment at end}$

At B at end:

$$y_B = y_{\max} = \frac{-M_B L^2}{2EI}$$

Between A and B:

$$y = \frac{-M_B x^2}{2EI}$$

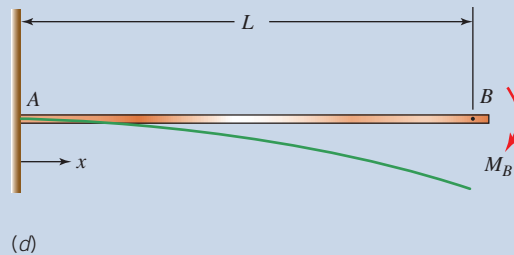

Source: *Engineering Data for Aluminum Structures* (Washington, DC: The Aluminum Association, 1986), pp. 63-77.

TABLE A14-3 Beam Diagrams and Beam-Deflection Formulas for Statically Indeterminate Beams

Deflections

At B at load:

$$y_B = \frac{-7}{768} \frac{PL^3}{EI}$$

 $y_{\max}$ is at $v = 0.447L$ at D :

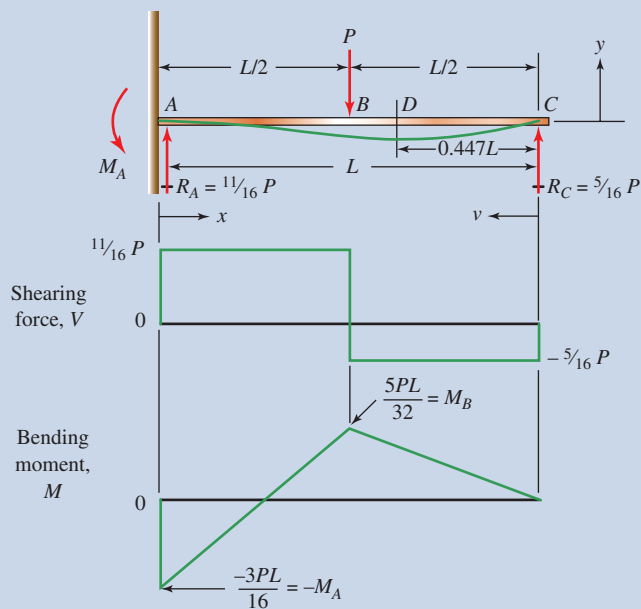
$$y_D = y_{\max} = \frac{-PL^3}{107EI}$$

Between A and B :

$$y = \frac{-Px^2}{96EI} (9L - 11x)$$

Between B and C :

$$y = \frac{-Pv}{96EI} (3L^2 - 5v^2)$$



(a)

Reactions

$$R_A = \frac{Pb}{2L^3} (3L^2 - b^2)$$

$$R_C = \frac{Pa^2}{2L^3} (b + 2L)$$

Moments

$$M_A = \frac{-Pab}{2L^2} (b + L)$$

$$M_B = \frac{Pa^2 b}{2L^3} (b + 2L)$$

Deflections

At B at load:

$$y_B = \frac{-Pa^3 b^2}{12EIL^3} (3L + b)$$

Between A and B :

$$y = \frac{-Px^2 b}{12EIL^3} (3C_1 - C_2 x)$$

$$C_1 = aL(L + b); C_2 = (L + a)(L + b) + aL$$

Between B and C :

$$y = \frac{-Pa^2 v}{12EIL^3} [3L^2 b - v^2(3L - a)]$$

(b)

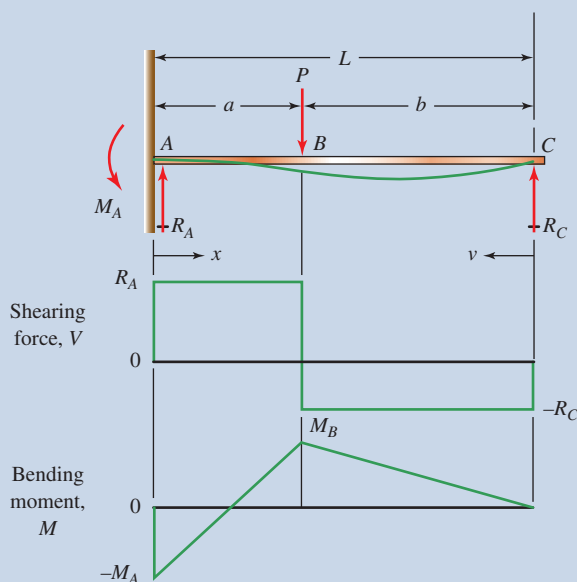


TABLE A14-3 (Continued)

Reactions

$$R_A = \frac{5}{8}W$$

$$R_B = \frac{3}{8}W$$

Moments

$$M_A = -0.125WL$$

$$M_E = 0.0703WL$$

Deflections

At C at $x = 0.579L$:

$$y_C = y_{\max} = \frac{-WL^3}{185EI}$$

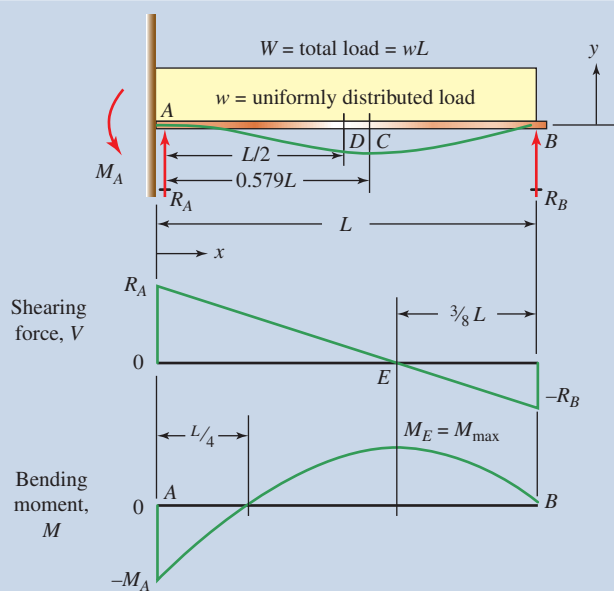
At D at center:

$$y_D = \frac{-WL^3}{192EI}$$

Between A and B:

$$y = \frac{-Wx^2(L-x)}{48EI}(3L-2x)$$

(c)



Reactions

$$R_A = \frac{-3Pa}{2L}$$

$$R_B = P\left(1 + \frac{3a}{2L}\right)$$

Moments

$$M_A = \frac{Pa}{2}$$

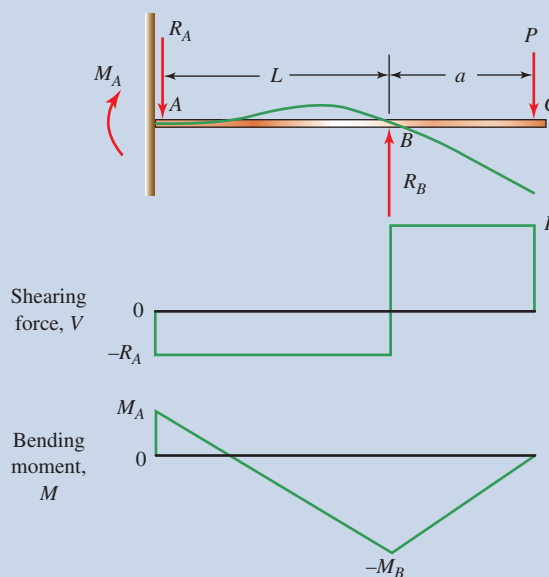
$$M_B = -Pa$$

Deflection

At C at end:

$$y_C = \frac{-PL^3}{EI}\left(\frac{a^2}{4L^2} + \frac{a^3}{3L^3}\right)$$

(d)



(Continued)

TABLE A14-3 (Continued)

Moments

$$M_B = PL/8; M_A = M_C = -PL/8$$

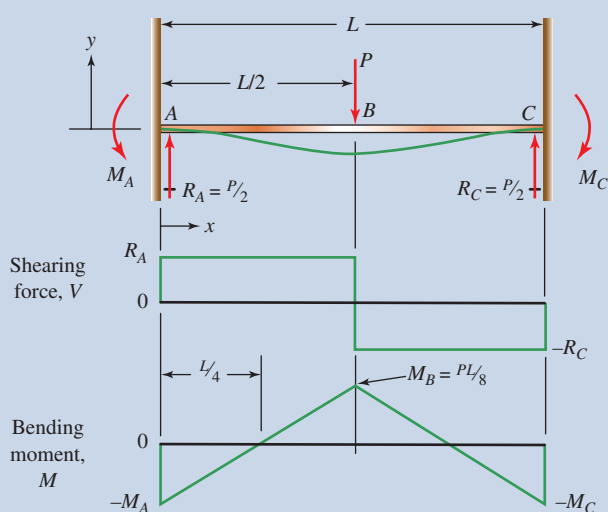
Deflections

At B at center:

$$y_B = y_{\max} = \frac{-PL^3}{192EI}$$

Between A and B:

$$y = \frac{-Px^2}{48EI}(3L - 4x)$$



(e)

Reactions

$$R_A = \frac{Pb^2}{L^3}(3a + b)$$

$$R_C = \frac{Pa^2}{L^3}(3b + a)$$

Moments

$$M_A = \frac{-Pab^2}{L^2}$$

$$M_B = \frac{2Pa^2b^2}{L^3}$$

$$M_C = \frac{-Pa^2b}{L^2}$$

Deflections

At B at load:

$$y_B = \frac{-Pa^3b^3}{3EIL^3}$$

At D at $x_1 = \frac{2aL}{3a + b}$

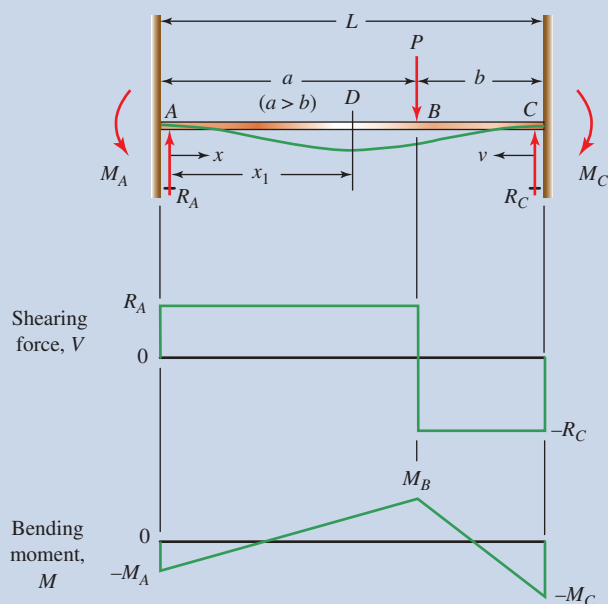
$$y_D = y_{\max} = \frac{-2Pa^3b^2}{3E(3a + b)^2}$$

Between A and B (longer segment):

$$y = \frac{-Px^2b^2}{6EIL^3}[2a(L - x) + L(a - x)]$$

Between B and C (shorter segment):

$$y = \frac{-Pv^2a^2}{6EIL^3}[2b(L - v) + L(b - v)]$$



(f)

TABLE A14-3 (Continued)

Moments

$$M_A = M_C = \frac{-WL}{12}$$

$$M_B = \frac{WL}{24}$$

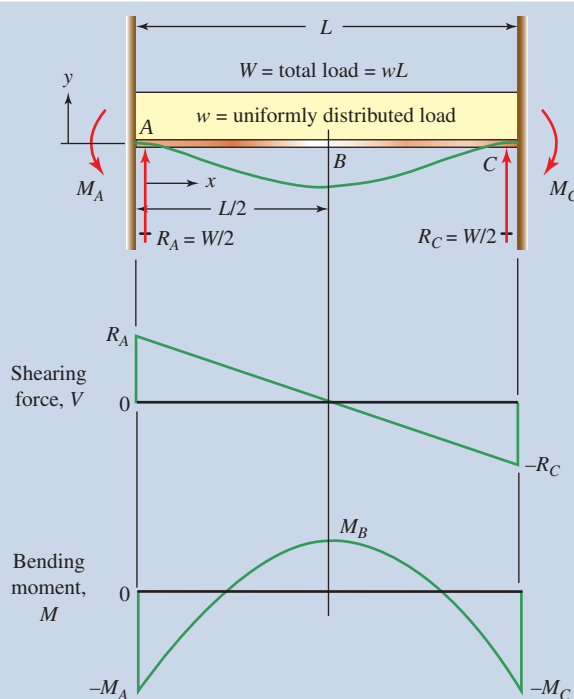
Deflections

At B at center:

$$y_B = y_{\max} = \frac{-WL^3}{384EI}$$

Between A and C:

$$y = \frac{-wx^2}{24EI}(L-x)^2$$



(g)

Reactions

$$R_A = R_C = \frac{3wL}{8}$$

$$R_B = 1.25wL$$

Shearing forces

$$V_A = V_C = R_A = R_C = \frac{3wL}{8}$$

$$V_B = \frac{5wL}{8}$$

Moments

$$M_D = M_E = 0.0703wL^2$$

$$M_B = -0.125wL^2$$

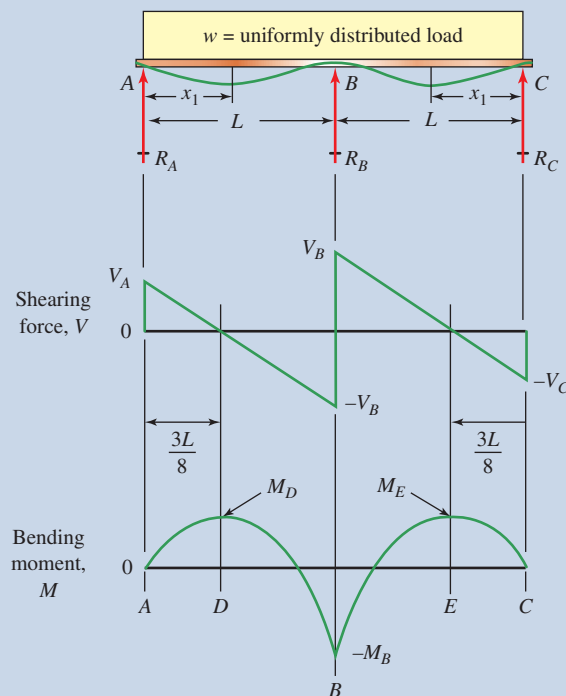
Deflections

At $x_1 = 0.4215L$ from A or C:

$$y_{\max} = \frac{-wL^4}{185EI}$$

Between A and B:

$$y = \frac{-w}{48EI}(L^3x - 3Lx^3 + 2x^4)$$



(h)

(Continued)

TABLE A14-3 (Continued)

Reactions

$$R_A = R_D = 0.4wL$$

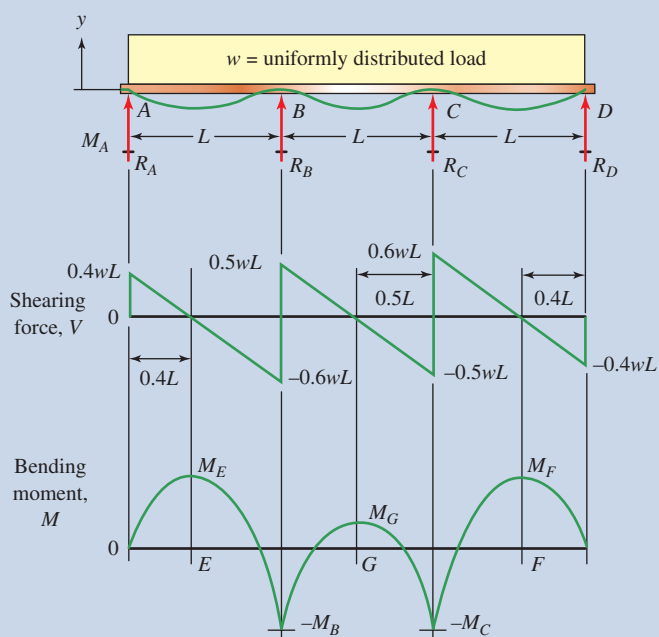
$$R_B = R_C = 1.10wL$$

Moments

$$M_E = M_F = 0.08wL^2$$

$$M_B = M_C = -0.10wL^2 = M_{\max}$$

$$M_G = 0.025wL^2$$



(i)

Reactions

$$R_A = R_E = 0.393wL$$

$$R_B = R_D = 1.143wL$$

$$R_C = 0.928wL$$

Shearing forces

$$V_A = +0.393wL$$

$$-V_B = -0.607wL$$

$$+V_B = +0.536wL$$

$$-V_C = -0.464wL$$

$$+V_C = +0.464wL$$

$$-V_D = -0.536wL$$

$$+V_D = +0.607wL$$

$$-V_E = -0.393wL$$

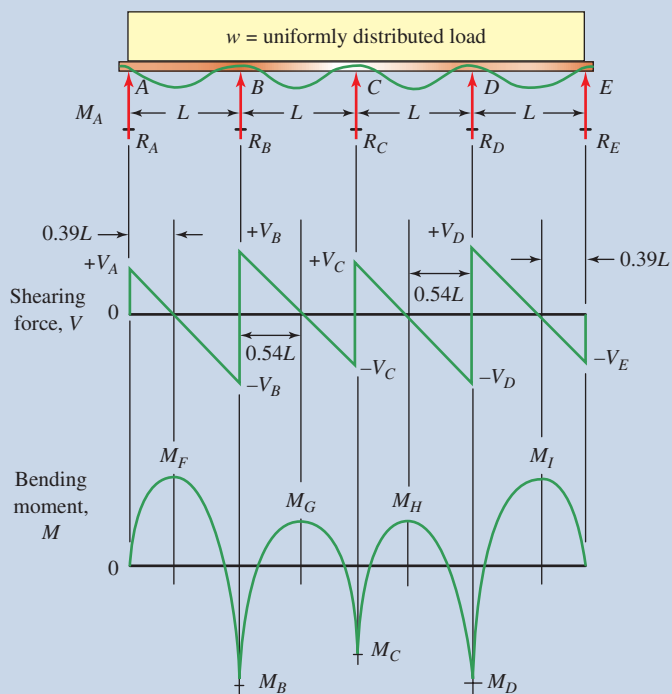
Moments

$$M_B = M_D = -0.1071wL^2 = M_{\max}$$

$$M_F = M_I = 0.0772wL^2$$

$$M_C = -0.0714wL^2$$

$$M_G = M_H = 0.0364wL^2$$



(j)

Source: Engineering Data for Aluminum Structures (Washington, DC: The Aluminum Association, 1986), pp. 63-77.

APPENDIX 15 Commercially Available Shapes Used for Load-Carrying Members

Table No.	Units	Description
15-1	U.S.	Angles: Steel and aluminum, equal and unequal legs, L-shapes, larger sizes: 2 in to 8 in
15-2	U.S.	Angles: Steel and aluminum, equal and unequal legs, smaller sizes: 0.50 in to 2.5 in
15-3	SI	Angles: Steel and aluminum, equal and unequal legs, 10 mm to 100 mm
15-4	U.S.	Channels: Steel and aluminum, American Standard, C-shapes, larger sizes: 3 in to 15 in depth
15-5	U.S.	Channels: Aluminum, smaller sizes: 0.5 in to 2.0 in depth
15-6	U.S.	Channels: Aluminum, Aluminum Association Standard shapes, larger sizes: 2 in to 12 in depth
15-7	SI	Channels: Aluminum, European standard shapes: 10 mm to 160 mm depth
15-8	SI	Channels: Steel, European standard shapes: 30 mm to 400 mm depth
15-9	U.S.	I-beam shapes: Steel wide-flange shapes, W-shapes: 4 in to 24 in depth
15-10	U.S.	I-beam shapes: Steel, American Standard, S-shapes: 3 in to 24 in depth
15-11	U.S.	I-beam shapes: Aluminum Association standard shapes: 3 in to 12 in depth
15-12	U.S.	I-beam shapes: Aluminum, small extruded shapes: 0.70 in to 2.11 in depth
15-13	SI	I-beam shapes: Steel, European standard shapes: 80 mm to 600 mm depth
15-14	U.S.	Hollow tubing: Steel, square and rectangular, standard structural HSS shapes: 2 in to 8 in depth
15-15	U.S.	Hollow tubing: Steel and aluminum, square and rectangular, smaller sizes: 0.375 in to 3.00 in depth
15-16	SI	Hollow tubing: Steel and aluminum, square and rectangular: 20 mm to 300 mm depth
15-17	U.S.	Pipe: Steel, American National Standard Schedule 40 and AISC standard: 1/8 in to 18 in sizes
15-18	U.S.	Mechanical tubing: Steel and aluminum: 0.50 in to 5.0 in outside diameters
15-19	SI	Mechanical tubing: Steel and aluminum: 10 mm to 150 mm outside diameters

Notes: Each table lists sample sizes of shapes commonly used for beam, column, or tension members.

Included are standard size designations, detailed dimensions, and section properties.

Section properties include cross-sectional area, moment of inertia, and section modulus.

For some sections, torsional moment of inertia, section modulus, and radius of gyration are also listed.

Numerous additional shapes are typically commercially available.

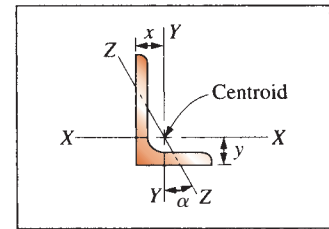
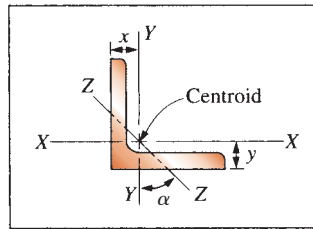
Internet sites for companies supplying each kind of shape are listed to aid in finding other sizes.

However, not all sites include the section properties data. See Appendix 1 for formulas for section properties.

Actual data listed obtained from a variety of sources.

Angles: Equal and unequal legs

Tables 15-1 to 15-3



U.S. Units

TABLE 15-1 Angles: Steel and Aluminum, Equal and Unequal Legs, L-Shapes, Larger Sizes: 2 in to 8 in

Designation	Area A (in ²)	Weight per foot (lb/ft)	*Section properties							
			Axis X-X			Axis Y-Y			Axis Z-Z	
			I_x (in ⁴)	S_x (in ³)	y (in)	I_y (in ⁴)	S_y (in ³)	x (in)	r (in)	α (deg)
L8×8×1	15.0	51.0	89.0	15.8	2.37	89.0	15.8	2.37	1.56	45.0
L8×8×1/2	7.75	26.4	48.6	8.36	2.19	48.6	8.36	2.19	1.59	45.0
L8×4×1	11.0	37.4	69.6	14.1	3.05	11.6	3.94	1.05	0.846	13.9
L8×4×1/2	5.75	19.6	38.5	7.49	2.86	6.74	2.15	0.859	0.865	14.9
L6×6×3/4	8.44	28.7	28.2	6.66	1.78	28.2	6.66	1.78	1.17	45.0
L6×6×3/8	4.36	14.9	15.4	3.53	1.64	15.4	3.53	1.64	1.19	45.0
L6×4×3/4	6.94	23.6	24.5	6.25	2.08	8.68	2.97	1.08	0.860	23.2
L6×4×3/8	3.61	12.3	13.5	3.32	1.94	4.90	1.60	0.941	0.877	24.0
L4×4×1/2	3.75	12.8	5.56	1.97	1.18	5.56	1.97	1.18	0.782	45.0
L4×4×1/4	1.94	6.6	3.04	1.05	1.09	3.04	1.05	1.09	0.795	45.0
L4×3×1/2	3.25	11.1	5.05	1.89	1.33	2.42	1.12	0.827	0.639	28.5
L4×3×1/4	1.69	5.8	2.77	1.00	1.24	1.36	0.599	0.896	0.651	29.2
L3×3×1/2	2.75	9.4	2.22	1.07	0.932	2.22	1.07	0.932	0.584	45.0
L3×3×1/4	1.44	4.9	1.24	0.577	0.842	1.24	0.577	0.842	0.592	45.0
L2×2×3/8	1.36	4.7	0.479	0.351	0.636	0.479	0.351	0.636	0.389	45.0
L2×2×1/4	0.938	3.19	0.348	0.247	0.592	0.348	0.247	0.592	0.391	45.0
L2×2×1/8	0.484	1.65	0.190	0.131	0.546	0.190	0.131	0.546	0.398	45.0

Note: I = moment of inertia; S = section modulus; r = radius of gyration.

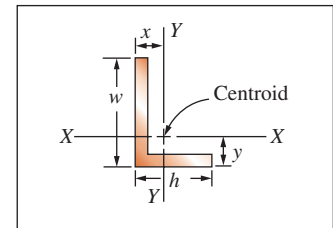
Example designation: L4×3×1/2.

4 = length of longer leg (in); 3 = length of shorter leg (in); 1/2 = thickness of legs (in).

Z-Z is axis of minimum moment of inertia (I) and radius of gyration (r).

Sources for data for additional sizes: Central Steel & Wire Co., multiple locations.

Earl M. Jorgensen Co., multiple locations.



U.S. Units

TABLE 15-2 Angles: Steel and Aluminum, Equal and Unequal Legs, Smaller Sizes: 0.50 in to 2.5 in

Ref.	Short side h (in)	Long side w (in)	Thickness t (in)	Area A (in ²)	**Weight/ft		*Section properties					
					Steel (lb/ft)	Aluminum (lb/ft)	Axis X-X			Axis Y-Y		
							I_x (in ⁴)	S_x (in ³)	y (in)	I_y (in ⁴)	S_y (in ³)	x (in)
a	1/2	1/2	1/8	0.109	0.372	0.131	0.00230	0.00698	0.170	0.00230	0.00698	0.170
b	5/8	5/8	1/8	0.141	0.478	0.169	0.00479	0.0113	0.201	0.00479	0.0113	0.201
c	7/8	7/8	1/8	0.203	0.691	0.244	0.01420	0.0233	0.264	0.0142	0.0233	0.264
d	5/8	1	1/8	0.188	0.637	0.225	0.00553	0.0121	0.167	0.0185	0.0286	0.354
e	3/4	1	1/8	0.203	0.691	0.244	0.00947	0.0174	0.207	0.0197	0.0295	0.332
f	1	1	1/8	0.234	0.797	0.281	0.0217	0.0309	0.296	0.0217	0.0309	0.296
g	1	1	1/4	0.438	1.487	0.525	0.0369	0.0558	0.339	0.0369	0.0558	0.339
h	1 1/4	1 1/4	1/8	0.297	1.009	0.356	0.0439	0.0493	0.359	0.0439	0.0493	0.359
i	1 1/4	1 1/4	1/4	0.563	1.912	0.675	0.0767	0.0905	0.403	0.0767	0.0905	0.403
j	7/8	13/8	1/8	0.266	0.903	0.319	0.0162	0.0247	0.217	0.0509	0.0560	0.467
k	1 1/4	1 1/2	3/16	0.480	1.633	0.577	0.0651	0.0726	0.353	0.1035	0.1013	0.478
l	1 1/2	1 1/2	1/8	0.359	1.222	0.431	0.0778	0.0721	0.421	0.0778	0.0721	0.421
m	1 1/2	1 1/2	1/4	0.688	2.337	0.825	0.1385	0.1340	0.466	0.139	0.1340	0.466
n	1 1/4	1 3/4	1/8	0.359	1.222	0.431	0.0486	0.0515	0.307	0.113	0.0943	0.557
o	1 3/4	1 3/4	1/8	0.422	1.434	0.506	0.1256	0.0992	0.484	0.126	0.0992	0.484
p	1 3/4	1 3/4	1/4	0.813	2.762	0.975	0.2272	0.1860	0.529	0.227	0.186	0.529
q	1 1/4	2	3/16	0.574	1.952	0.689	0.0707	0.0752	0.311	0.232	0.177	0.686
r	1 1/2	2	1/8	0.422	1.434	0.506	0.0847	0.0748	0.368	0.173	0.125	0.618
s	1 1/2	2	1/4	0.813	2.762	0.975	0.1515	0.139	0.413	0.316	0.236	0.663
t	1 1/2	2 1/2	3/16	0.715	2.430	0.858	0.1275	0.111	0.352	0.461	0.279	0.852

Notes: * I = moment of inertia; S = section modulus; See sketch for X-X and Y-Y axes and their locations x and y .

Values are for perfectly square corners. Some vendors apply radii to inside and/or outside corners.

Numerous additional sizes available. Consult vendors.

** Using density of steel = 0.283 lb_m/in³; density of aluminum = 0.100 lb_m/in³. (May range from 0.095 lb_m/in³ to 0.102 lb_m/in³.)

Some sizes are also available in stainless steel. Check with vendor.

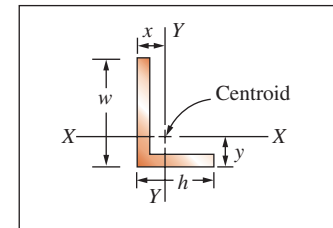
Sources for data for additional sizes: Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations.

Central Steel & Wire Co., multiple locations.

OnlineMetals.com, Seattle, WA.

Paramount Extrusions, Co., Paramount, CA.

Metals Depot, Winchester, KY.



SI Units

TABLE 15-3 Angles: Steel and Aluminum, Equal and Unequal Legs, 10 mm to 100 mm

Ref.	Short side <i>h</i> (mm)	Long side <i>w</i> (mm)	Thickness <i>t</i> (mm)	Area <i>A</i> (mm ²)	**Weight/m		*Section properties					
					Steel (N/m)	Aluminum (N/m)	Axis X-X			Axis Y-Y		
							<i>I_x</i> (mm ⁴)	<i>S_x</i> (mm ³)	<i>y</i> (mm)	<i>I_y</i> (mm ⁴)	<i>S_y</i> (mm ³)	<i>x</i> (mm)
a	10	15	3	66.0	4.973	1.79	477	69.1	3.09	1375	146	5.59
b	10	20	3	81.0	6.103	2.20	520	72.1	2.80	3140	257	7.80
c	15	20	3	96.0	7.233	2.61	1743	163	4.31	3653	277	6.81
d	15	25	3	111	8.363	3.01	1856	168	3.93	6876	428	8.93
e	15	30	3	126	9.493	3.42	1946	171	3.64	1.146E+04	608	11.1
f	20	20	3	111	8.363	3.01	4030	290	6.09	4030	290	6.09
g	20	30	3	141	10.623	3.83	4551	306	5.12	1.272E+04	640	10.1
h	20	40	3	171	12.883	4.64	4897	316	4.48	2.829E+04	1109	14.5
i	25	25	3	141	10.623	3.83	8204	465	7.35	8.204E+03	465	7.35
j	30	30	5	275	20.719	7.47	2.216E+04	1072	9.32	2.216E+04	1072	9.32
k	30	50	4	304	22.904	8.25	2.152E+04	941	7.13	7.800E+04	2373	17.1
l	40	40	5	375	28.253	10.18	5.561E+04	1974	11.8	5.561E+04	1974	11.8
m	40	60	5	475	35.787	12.90	6.270E+04	2081	9.87	1.740E+05	4334	19.9
n	40	80	6	684	51.533	18.57	7.836E+04	2525	8.96	4.526E+05	8868	29.0
o	50	50	5	475	35.787	12.90	1.125E+05	3155	14.3	1.125E+05	3155	14.3
p	50	75	5	600	45.205	16.29	1.266E+05	3322	11.9	3.485E+05	6884	24.4
q	50	100	6	864	65.095	23.46	1.590E+05	4039	10.6	9.058E+05	14073	35.6
r	60	60	6	684	51.533	18.57	2.333E+05	5452	17.2	2.333E+05	5452	17.2
s	75	75	6	864	65.095	23.46	4.688E+05	8677	21.0	4.688E+05	8677	21.0
t	100	100	10	1900	143.148	51.59	1.800E+06	25240	28.7	1.800E+06	25240	28.7

Notes: **I* = moment of inertia; *S* = section modulus; See sketch for X-X and Y-Y axes and their locations *x* and *y*.

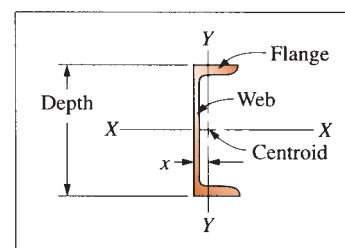
Values are for perfectly square corners. Some vendors apply radii to inside and/or outside corners.

**Using density of steel = 7680 kg/m³; density of aluminum = 2768 kg/m³. (May range from 2635 kg/m³ to 2829 kg/m³.)

Source for data for additional sizes: Parker Steel Company—Metric Sized Metals, Toledo, Ohio.

Channels

Tables 15–4 to 15–8



U.S. Units

TABLE 15–4 Channels: Steel and Aluminum, American Standard C-Shapes, Larger Sizes: 3 in to 15 in Depth

Designation	Area A (in ²)	Depth (in)	Web thickness (in)	*Section properties						
				Flange		Axis X–X		Axis Y–Y		
				Width (in)	Average thickness (in)	I_x (in ⁴)	S_x (in ³)	I_y (in ⁴)	S_y (in ³)	x (in)
C15×50	14.7	15.00	0.716	3.716	0.650	404	53.8	11.0	3.78	0.798
C15×40	11.8	15.00	0.520	3.520	0.650	349	46.5	9.23	3.37	0.777
C12×30	8.82	12.00	0.510	3.170	0.501	162	27.0	5.14	2.06	0.674
C12×25	7.35	12.00	0.387	3.047	0.501	144	24.1	4.47	1.88	0.674
C10×30	8.82	10.00	0.673	3.033	0.436	103	20.7	3.94	1.65	0.649
C10×20	5.88	10.00	0.379	2.739	0.436	78.9	15.8	2.81	1.32	0.606
C9×20	5.88	9.00	0.448	2.648	0.413	60.9	13.5	2.42	1.17	0.583
C9×15	4.41	9.00	0.285	2.485	0.413	51.0	11.3	1.93	1.01	0.586
C8×18.75	5.51	8.00	0.487	2.527	0.390	44.0	11.0	1.98	1.01	0.565
C8×11.5	3.38	8.00	0.220	2.260	0.390	32.6	8.14	1.32	0.781	0.571
C6×13	3.83	6.00	0.437	2.157	0.343	17.4	5.80	1.05	0.642	0.514
C6×8.2	2.40	6.00	0.200	1.920	0.343	13.1	4.38	0.693	0.492	0.511
C5×9	2.64	5.00	0.325	1.885	0.320	8.90	3.56	0.632	0.450	0.478
C5×6.7	1.97	5.00	0.190	1.750	0.320	7.49	3.00	0.479	0.378	0.484
C4×7.25	2.13	4.00	0.321	1.721	0.296	4.59	2.29	0.433	0.343	0.459
C4×5.4	1.59	4.00	0.184	1.584	0.296	3.85	1.93	0.319	0.283	0.457
C3×6	1.76	3.00	0.356	1.596	0.273	2.07	1.38	0.305	0.268	0.455
C3×4.1	1.21	3.00	0.170	1.410	0.273	1.66	1.10	0.197	0.202	0.436

Note: Example designation: C15×50.

15 = depth (in); 50 = weight per unit length (lb/ft).

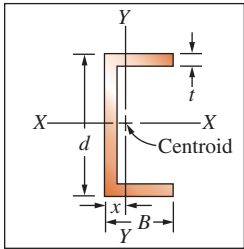
* I = moment of inertia; S = section modulus.

Sources for data for additional sizes: Central Steel & Wire Co., multiple locations.

Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations.

Metals Depot, Winchester, KY.

OnlineMetals.com, Seattle, WA.



U.S. Units

TABLE 15-5 Channels: Aluminum, Smaller Sizes: 0.5 in to 2.0 in Depth

Ref.	Depth <i>d</i> (in)	Flange width <i>B</i> (in)	Thickness <i>t</i> (in)	**Weight/ft (lb/ft)	Area <i>A</i> (in ²)	*Section properties				
						Axis X-X		Axis Y-Y		
						<i>I_x</i> (in ⁴)	<i>S_x</i> (in ³)	<i>I_y</i> (in ⁴)	<i>S_y</i> (in ³)	<i>x</i> (in)
a	0.500	0.375	0.125	0.150	0.1250	0.00358	0.0143	0.001506	0.00688	0.156
b	0.500	0.500	0.094	0.148	0.123	0.00418	0.0167	0.00290	0.00970	0.202
c	0.500	0.750	0.125	0.263	0.219	0.0070	0.0280	0.01145	0.0273	0.330
d	0.625	0.625	0.125	0.244	0.203	0.0105	0.0337	0.00740	0.01999	0.255
e	0.750	0.375	0.125	0.188	0.156	0.0106	0.0282	0.001766	0.0074	0.138
f	0.750	0.750	0.125	0.300	0.250	0.0199	0.0530	0.01345	0.02968	0.297
g	1.000	0.500	0.125	0.263	0.219	0.0285	0.057	0.00461	0.0140	0.170
h	1.000	1.000	0.125	0.413	0.344	0.0526	0.105	0.03401	0.0549	0.381
i	1.250	0.500	0.125	0.300	0.250	0.0501	0.080	0.00496	0.0144	0.156
j	1.250	1.250	0.125	0.525	0.438	0.1097	0.176	0.06910	0.0879	0.464
k	1.500	0.500	0.125	0.338	0.281	0.080	0.106	0.00525	0.0148	0.146
l	1.750	0.500	0.125	0.375	0.313	0.118	0.135	0.00549	0.0151	0.138
m	1.750	0.750	0.125	0.450	0.375	0.159	0.182	0.01819	0.0342	0.219
n	1.750	1.000	0.125	0.525	0.438	0.201	0.229	0.04159	0.0605	0.313
o	2.000	0.500	0.125	0.413	0.344	0.166	0.166	0.00569	0.01539	0.131

Notes: **I* = moment of inertia; *S* = section modulus; See sketch for X-X and Y-Y axes and the location *x* for the axis Y-Y.

Section properties computed assuming square corners and constant thickness of web and flanges.

** Using density for aluminum of 0.100 lb_m/in³. May range from 0.095 lb_m/in³ to 0.102 lb_m/in³ for different alloys.

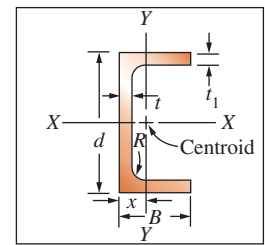
Sources for data for additional sizes: Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations.

Central Steel & Wire Co., multiple locations.

OnlineMetals.com, Seattle, WA.

Paramount Extrusions, Co., Paramount, CA.

Metals Depot, Winchester, KY.



U.S. Units

TABLE 15-6 Channels: Aluminum, Aluminum Association Standard Shapes, Larger Sizes: 2 in to 12 in Depth

Size		¹ Section properties											
Depth <i>d</i> (in)	Width <i>B</i> (in)	Area <i>A</i> (in ²)	Weight ² (lb/ft)	Flange thickness <i>t</i> ₁ (in)	Web thickness <i>t</i> (in)	Fillet radius <i>R</i> (in)	Axis <i>X-X</i>			Axis <i>Y-Y</i>			
							<i>I</i> _{<i>x</i>} (in ⁴)	<i>S</i> _{<i>x</i>} (in ³)	<i>r</i> _{<i>x</i>} (in)	<i>I</i> _{<i>y</i>} (in ⁴)	<i>S</i> _{<i>y</i>} (in ³)	<i>r</i> _{<i>y</i>} (in)	<i>x</i> (in)
2.00	1.00	0.491	0.577	0.13	0.13	0.10	0.288	0.288	0.766	0.045	0.064	0.303	0.298
2.00	1.25	0.911	1.071	0.26	0.17	0.15	0.546	0.546	0.774	0.139	0.178	0.391	0.471
3.00	1.50	0.965	1.135	0.20	0.13	0.25	1.41	0.94	1.21	0.22	0.22	0.47	0.49
3.00	1.75	1.358	1.597	0.26	0.17	0.25	1.97	1.31	1.20	0.42	0.37	0.55	0.62
4.00	2.00	1.478	1.738	0.23	0.15	0.25	3.91	1.95	1.63	0.60	0.45	0.64	0.65
4.00	2.25	1.982	2.331	0.29	0.19	0.25	5.21	2.60	1.62	1.02	0.69	0.72	0.78
5.00	2.25	1.881	2.212	0.26	0.15	0.30	7.88	3.15	2.05	0.98	0.64	0.72	0.73
5.00	2.75	2.627	3.089	0.32	0.19	0.30	11.14	4.45	2.06	2.05	1.14	0.88	0.95
6.00	2.50	2.410	2.834	0.29	0.17	0.30	14.35	4.78	2.44	1.53	0.90	0.80	0.79
6.00	3.25	3.427	4.030	0.35	0.21	0.30	21.04	7.01	2.48	3.76	1.76	1.05	1.12
7.00	2.75	2.725	3.205	0.29	0.17	0.30	22.09	6.31	2.85	2.10	1.10	0.88	0.84
7.00	3.50	4.009	4.715	0.38	0.21	0.30	33.79	9.65	2.90	5.13	2.23	1.13	1.20
8.00	3.00	3.526	4.147	0.35	0.19	0.30	37.40	9.35	3.26	3.25	1.57	0.96	0.93
8.00	3.75	4.923	5.789	0.41	0.25	0.35	52.69	13.17	3.27	7.13	2.82	1.20	1.22
9.00	3.25	4.237	4.983	0.35	0.23	0.35	54.41	12.09	3.58	4.40	1.89	1.02	0.93
9.00	4.00	5.927	6.970	0.44	0.29	0.35	78.31	17.40	3.63	9.61	3.49	1.27	1.25
10.00	3.50	5.218	6.136	0.41	0.25	0.35	83.22	16.64	3.99	6.33	2.56	1.10	1.02
10.00	4.25	7.109	8.360	0.50	0.31	0.40	116.15	23.23	4.04	13.02	4.47	1.35	1.34
12.00	4.00	7.036	8.274	0.47	0.29	0.40	159.76	26.63	4.77	11.03	3.86	1.25	1.14
12.00	5.00	10.053	11.822	0.62	0.35	0.45	239.69	39.95	4.88	25.74	7.60	1.60	1.61

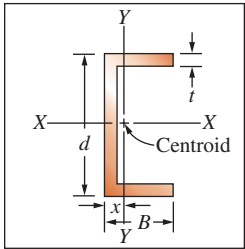
Notes: ¹*I* = moment of inertia; *S* = section modulus; *r* = radius of gyration.

²Weights per foot are based on nominal dimensions and a density of 0.098 lb/in³, which is the density of alloy 6061. (May Range from 0.095 lb_m/in³ to 0.102 lb_m/in³.)

Sources for data for additional sizes: Aluminum Association, *Aluminum Standards and Data*, 11th ed., Washington, DC, © 1993, p. 187.

Metals Depot, Winchester, KY.

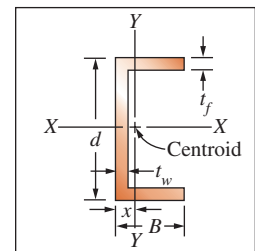
OnlineMetals.com, Seattle, WA.



SI Units

TABLE 15-7 Channels: Aluminum, European Standard Shapes: 10 mm to 160 mm Depth										
Ref.	Depth <i>d</i> (mm)	Flange width <i>B</i> (mm)	Thickness <i>t</i> (mm)	² Weight/m (N/m)	Area <i>A</i> (mm ²)	¹ Section properties				
						Axis X-X		Axis Y-Y		
						<i>I_x</i> (mm ⁴)	<i>S_x</i> (mm ³)	<i>I_y</i> (mm ⁴)	<i>S_y</i> (mm ³)	<i>x</i> (mm)
a	10	10	2	1.10	40.5	590.4	118.1	392	64.32	3.90
b	12	12	2	1.74	64.0	1301	216.9	881	121.6	4.75
c	15	15	2	2.23	82.0	2777	370.2	1812	196.1	5.76
d	20	20	3	4.40	162	9446	944.6	6279	514.5	7.80
e	25	20	2	3.31	122	1.215E+04	972.0	4911	375.0	6.90
f	25	25	3	5.62	207	1.998E+04	1598	1.285E+04	827.7	9.47
g	35	35	3	8.06	297	6.001E+04	3429	3.725E+04	1679	12.81
h	40	30	4	9.99	368	8.900E+04	4450	3.228E+04	1653	10.48
i	50	30	4	11.08	408	1.520E+05	6079	3.493E+04	1716	9.65
j	50	40	4	13.25	488	1.944E+05	7776	7.858E+04	3000	13.80
k	60	40	4	14.34	528	2.982E+05	9939	8.379E+04	3093	12.91
i	80	40	4	16.51	608	5.869E+05	1.467E+04	9.216E+04	3231	11.47
m	100	50	5	25.79	950	1.433E+06	2.866E+04	2.250E+05	6310	14.34
n	125	80	8	58.43	2152	5.251E+06	8.401E+04	1.360E+06	24905	25.41
o	160	80	10	81.46	3000	1.130E+07	1.413E+05	1.780E+06	31592	23.67

Notes: ¹*I* = moment of inertia; *S* = section modulus; See sketch for X-X and Y-Y axes and the location *x* for the axis Y-Y.
Section properties computed assuming square corners and constant thickness of web and flanges.
Some producers use filleted corners and tapered flanges. Adjustments to computed values may be needed for precision.
²Using density of aluminum = 2768 kg/m³. (May range from 2635 kg/m³ to 2829 kg/m³.)
Sources for data for additional sizes: Parker Steet Company—Metric Sized Metals, Toledo, Ohio.



SI Units

TABLE 15-8 Channels: Steel, European Standard Shapes: 30 mm to 400 mm Depth

Ref.	Depth d (mm)						¹ Section properties				
		Flange		Web	² Weight/m (N/m)	Area A (mm ²)	Axis X-X		Axis Y-Y		
		Width B (mm)	Thickness t_f (mm)	thickness t_w (mm)			I_x (mm ⁴)	S_x (mm ³)	I_y (mm ⁴)	S_y (mm ³)	x (mm)
a	30	15	4.5	4.0	16.5	219.0	2.399E+04	1.599	4.210E+03	438	5.39
b	30	33	7.0	5.0	40.8	542.0	5.558E+04	3706	5.546E+04	2987	14.43
c	40	20	5.5	5.0	27.5	365.0	7.292E+04	3646	1.255E+04	967	7.02
d	40	35	7.0	5.0	46.7	620.0	1.192E+05	5958	7.341E+04	3556	14.35
e	50	25	6.0	5.0	36.9	490.0	1.538E+05	6150	2.765E+04	1688	8.62
f	50	38	7.0	5.0	53.6	712.0	2.198E+05	8793	1.010E+05	4359	14.83
g	60	30	6.0	6.0	48.8	648.0	3.188E+05	1.063E+04	5.090E+04	2503	9.67
h	80	45	8.0	6.0	83.2	1104.0	8.981E+05	2.245E+04	2.179E+05	7441	15.72
i	100	50	8.5	6.0	101.6	1348.0	1.668E+06	3.336E+04	3.306E+05	9.978E+03	16.87
j	140	60	10.0	7.0	153.7	2040.0	4.885E+06	6.979E+04	7.104E+05	1.736E+04	19.09
k	200	75	11.5	8.5	243.3	3229.5	1.604E+07	1.604E+05	1.706E+06	3.220E+04	22.01
l	260	90	14.0	10.0	364.7	4840.0	3.966E+07	3.051E+05	3.653E+06	5.692E+04	25.83
m	300	100	16.0	10.0	443.0	5880.0	6.036E+07	4.024E+05	5.642E+06	8.002E+04	29.49
n	350	100	16.0	14.0	576.5	7652.0	1.180E+08	6.744E+05	6.182E+06	8.240E+04	24.98
o	400	110	18.0	14.0	682.3	9056.0	1.748E+08	8.742E+05	9.210E+06	1.123E+05	27.99

Notes: ¹ I = moment of inertia; S = section modulus; See sketch for X-X and Y-Y axes and the location x for the axis Y-Y.

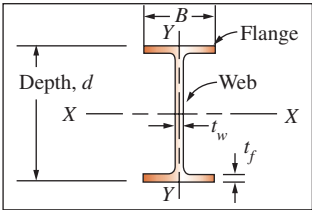
Section properties computed assuming square corners and constant thickness of web and flanges.

Some producers use filleted corners and tapered flanges. Adjustments to computed values may be needed for precision.

² Using density of steel = 7680 kg/m³.

Source for data for additional sizes: Parker Steel Company—Metric Sized Metals, Toledo, Ohio.

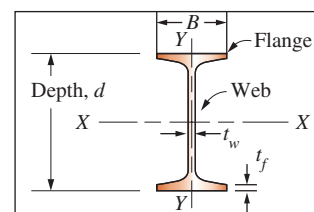
I-Beam Shapes
Tables 15–9 to 15–13



U.S. Units

TABLE 15–9 I-Beam Shapes: Steel Wide Flange Shapes, W-Shapes: 4 in to 24 in Depth									
Designation	Area <i>A</i> (in ²)	Depth <i>d</i> (in)	Web thickness <i>t_w</i> (in)	Flange		*Section properties			
				Width, <i>B</i> (in)	Thickness, <i>t_f</i> (in)	Axis <i>X–X</i>		Axis <i>Y–Y</i>	
						<i>I_x</i> (in ⁴)	<i>S_x</i> (in ³)	<i>I_y</i> (in ⁴)	<i>S_y</i> (in ³)
W24×76	22.4	23.92	0.440	8.990	0.680	2100	176	82.5	18.4
W24×68	20.1	23.73	0.415	8.965	0.585	1830	154	70.4	15.7
W21×73	21.5	21.24	0.455	8.295	0.740	1600	151	70.6	17.0
W21×57	16.7	21.06	0.405	6.555	0.650	1170	111	30.6	9.35
W18×55	16.2	18.11	0.390	7.530	0.630	890	98.3	44.9	11.9
W18×40	11.8	17.90	0.315	6.015	0.525	612	68.4	19.1	6.35
W14×43	12.6	13.66	0.305	7.995	0.530	428	62.7	45.2	11.3
W14×26	7.69	13.91	0.255	5.025	0.420	245	35.3	8.91	3.54
W12×30	8.79	12.34	0.260	6.520	0.440	238	38.6	20.3	6.24
W12×16	4.71	11.99	0.220	3.990	0.265	103	17.1	2.82	1.41
W10×15	4.41	9.99	0.230	4.000	0.270	69.8	13.8	2.89	1.45
W10×12	3.54	9.87	0.190	3.960	0.210	53.8	10.9	2.18	1.10
W8×15	4.44	8.11	0.245	4.015	0.315	48.0	11.8	3.41	1.70
W8×10	2.96	7.89	0.170	3.940	0.205	30.8	7.81	2.09	1.06
W6×15	4.43	5.99	0.230	5.990	0.260	29.1	9.72	9.32	3.11
W6×12	3.55	6.03	0.230	4.000	0.280	22.1	7.31	2.99	1.50
W5×19	5.54	5.15	0.270	5.030	0.430	26.2	10.2	9.13	3.63
W5×16	4.68	5.01	0.240	5.000	0.360	21.3	8.51	7.51	3.00
W4×13	3.83	4.16	0.280	4.060	0.345	11.3	5.46	3.86	1.90

Note: Example designation: W14×43.
14 = nominal depth (in); 43 = weight per unit length (lb/ft).
**I* = moment of inertia; *S* = section modulus.
Sources for data for additional sizes: Central Steel & Wire Co., multiple locations.
Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations.



U.S. Units

TABLE 15-10 I-Beam Shapes: Steel, American Standard Shapes, S-Shapes: 3 in to 24 in Depth

Designation	Area A (in ²)	Depth d (in)	Web thickness t_w (in)	Flange		*Section properties			
				Width B (in)	Average thickness t_f (in)	Axis X-X		Axis Y-Y	
						I_x (in ⁴)	S_x (in ³)	I_y (in ⁴)	S_y (in ³)
S24×90	26.5	24.00	0.625	7.125	0.870	2250	187	44.9	12.6
S20×96	28.2	20.30	0.800	7.200	0.920	1670	165	50.2	13.9
S20×75	22.0	20.00	0.635	6.385	0.795	1280	128	29.8	9.32
S20×66	19.4	20.00	0.505	6.255	0.795	1190	119	27.7	8.85
S18×70	20.6	18.00	0.711	6.251	0.691	926	103	24.1	7.72
S15×50	14.7	15.00	0.550	5.640	0.622	486	64.8	15.7	5.57
S12×50	14.7	12.00	0.687	5.477	0.659	305	50.8	15.7	5.74
S12×35	10.3	12.00	0.428	5.078	0.544	229	38.2	9.87	3.89
S10×35	10.3	10.00	0.594	4.944	0.491	147	29.4	8.36	3.38
S10×25.4	7.46	10.00	0.311	4.661	0.491	124	24.7	6.79	2.91
S8×23	6.77	8.00	0.441	4.171	0.426	64.9	16.2	4.31	2.07
S8×18.4	5.41	8.00	0.271	4.001	0.426	57.6	14.4	3.73	1.86
S7×20	5.88	7.00	0.450	3.860	0.392	42.4	12.1	3.17	1.64
S6×12.5	3.67	6.00	0.232	3.332	0.359	22.1	7.37	1.82	1.09
S5×10	2.94	5.00	0.214	3.004	0.326	12.3	4.92	1.22	0.809
S4×7.7	2.26	4.00	0.193	2.663	0.293	6.08	3.04	0.764	0.574
S3×5.7	1.67	3.00	0.170	2.330	0.260	2.52	1.68	0.455	0.390

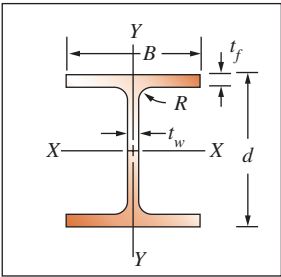
Note: Example designation: S10×35.

10 = nominal depth (in); 35 = weight per unit length (lb/ft).

* I = moment of inertia; S = section modulus.

Sources for data for additional sizes: Central Steel & Wire Co., multiple locations.

Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations.



U.S. Units

TABLE 15-11 I-Beam shapes: Aluminum, Aluminum Association Standard Shapes, Larger Sizes: 3 in to 12 in Depth												
Size		¹ Section properties										
Dept <i>d</i> (in)	Width <i>B</i> (in)	Area <i>A</i> (in ²)	² Weight (lb/ft)	Flange thickness <i>t_f</i> (in)	Web thickness <i>t_w</i> (in)	Fillet radius <i>R</i> (in)	Axis <i>X-X</i>			Axis <i>Y-Y</i>		
							<i>I_x</i> (in ⁴)	<i>S_x</i> (in ³)	<i>r_x</i> (in)	<i>I_y</i> (in ⁴)	<i>S_y</i> (in ³)	<i>r_y</i> (in)
3.00	2.50	1.392	1.637	0.20	0.13	0.25	2.24	1.49	1.27	0.52	0.42	0.61
3.00	2.50	1.726	2.030	0.26	0.15	0.25	2.71	1.81	1.25	0.68	0.54	0.63
4.00	3.00	1.965	2.311	0.23	0.15	0.25	5.62	2.81	1.69	1.04	0.69	0.73
4.00	3.00	2.375	2.793	0.29	0.17	0.25	6.71	3.36	1.68	1.31	0.87	0.74
5.00	3.50	3.146	3.700	0.32	0.19	0.30	13.94	5.58	2.11	2.29	1.31	0.85
6.00	4.00	3.427	4.030	0.29	0.19	0.30	21.99	7.33	2.53	3.10	1.55	0.95
6.00	4.00	3.990	4.692	0.35	0.21	0.30	25.50	8.50	2.53	3.74	1.87	0.97
7.00	4.50	4.932	5.800	0.38	0.23	0.30	42.89	12.25	2.95	5.78	2.57	1.08
8.00	5.00	5.256	6.181	0.35	0.23	0.30	59.69	14.92	3.37	7.30	2.92	1.18
8.00	5.00	5.972	7.023	0.41	0.25	0.30	67.78	16.94	3.37	8.55	3.42	1.20
9.00	5.50	7.110	8.361	0.44	0.27	0.30	102.02	22.67	3.79	12.22	4.44	1.31
10.00	6.00	7.352	8.646	0.41	0.25	0.40	132.09	26.42	4.24	14.78	4.93	1.42
10.00	6.00	8.747	10.286	0.50	0.29	0.40	155.79	31.16	4.22	18.03	6.01	1.44
12.00	7.00	9.925	11.672	0.47	0.29	0.40	255.57	42.60	5.07	26.90	7.69	1.65
12.00	7.00	12.153	14.292	0.62	0.31	0.40	317.33	52.89	5.11	35.48	10.14	1.71

Notes: ¹ Weights per foot are based on nominal dimensions and a density of 0.098 lb/in³, which is the density of alloy 6061.

² Areas listed are based on nominal dimensions.

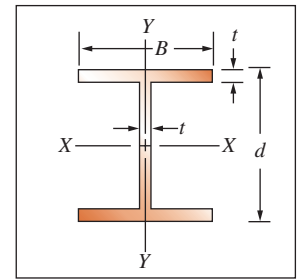
I = moment of inertia; *S* = section modulus; *r* = radius of gyration.

Sources for data for additional sizes: Aluminum Association, *Aluminum Standards and Data*, 11th ed., Washington, DC, © 1993, p. 187.

Central Steel & Wire Co., multiple locations.

Reliance Steel & Aluminum Co./Earl M. Jorgensen Co., multiple locations.

OnlineMetals.com, Seattle, WA.



U.S. Units

TABLE 15-12 I-Beam Shapes: Aluminum, Small Extruded Shapes: 0.70 in to 2.11 in Depth

Ref.	Depth d (in)	Flange width B (in)	Thickness t (in)	² Weight/ft (lb/ft)	Area A (in ²)	¹ Section properties					
						Axis $X-X$			Axis $Y-Y$		
						I_x (in ⁴)	S_x (in ³)	r_x (in)	I_y (in ⁴)	S_y (in ³)	r_y (in)
a	0.700	1.060	0.050	0.160	0.136	0.0121	0.0346	0.299	0.00993	0.0187	0.270
b	1.040	1.074	0.050	0.185	0.154	0.0298	0.0573	0.439	0.0103	0.0192	0.259
c	1.950	1.000	0.050	0.230	0.193	0.117	0.120	0.778	0.00835	0.0167	0.208
d	1.120	1.724	0.050	0.268	0.223	0.0538	0.0961	0.491	0.0427	0.0495	0.437
e	1.034	1.500	0.062	0.290	0.242	0.0479	0.0926	0.444	0.0349	0.0465	0.379
f	2.110	1.500	0.055	0.329	0.275	0.211	0.200	0.876	0.0310	0.0413	0.336
g	0.876	1.500	0.188	0.790	0.658	0.0704	0.161	0.327	0.106	0.141	0.401
h	1.000	2.375	0.250	1.575	1.313	0.176	0.352	0.366	0.559	0.471	0.653

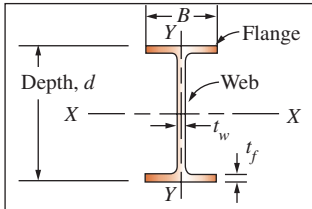
Notes: These shapes are designed for special applications and not made to common dimensions. Calculations assume square corners.

¹ I = moment of inertia; S = section modulus; r = radius of gyration, used for column analysis; See sketch for $X-X$ and $Y-Y$ axes.

²Using density of aluminum = 0.100 lb_m/in³. (May range from 0.095 lb_m/in³ to 0.102 lb_m/in³.)

Sources for data for additional sizes: Paramount Extrusions, Co., Paramount, CA.

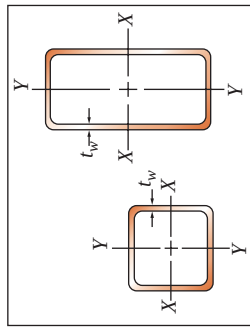
Metals Depot, Winchester, KY.



SI Units

TABLE 15-13 I-Beam Shapes: Steel, European Standard (² IPE) Shapes: 80 mm to 600 mm Depth															
							¹ Section properties								
							Thickness			Axis X-X			Axis Y-Y		
							Flange width <i>B</i> (mm)	Flange <i>t_f</i> (mm)	Web <i>t_w</i> (mm)	<i>I_x</i> (mm ⁴)	<i>S_x</i> (mm ³)	<i>r_x</i> (mm)	<i>I_y</i> (mm ⁴)	<i>S_y</i> (mm ³)	<i>r_y</i> (mm)
Ref.	Depth <i>d</i> (mm)				³ Weight/m (N/m)	Area <i>A</i> (mm ²)									
a	80	46	5.2	3.8	57.6	764	8.014E+05	2.003E+04	32.4	8.487E+04	3.690E+03	10.54			
b	100	55	5.7	4.1	77.8	1032	1.710E+06	3.420E+04	40.7	1.591E+05	5.786E+03	12.4			
c	120	64	6.3	4.4	99.5	1321	3.177E+06	5.296E+04	49.0	2.766E+05	8.644E+03	14.5			
d	140	73	6.9	4.7	123.8	1643	5.412E+06	7.732E+04	57.4	4.491E+05	1.230E+04	16.5			
e	160	82	7.4	5.0	151.4	2009	8.693E+06	1.087E+05	65.8	6.829E+05	1.666E+04	18.4			
f	180	91	8.0	5.3	180.4	2395	1.317E+07	1.463E+05	74.2	1.008E+06	2.216E+04	20.5			
g	200	100	8.5	5.6	214.6	2848	1.943E+07	1.943E+05	82.6	1.423E+06	2.846E+04	22.4			
h	220	110	9.2	5.9	251.4	3337	2.772E+07	2.520E+05	91.1	2.048E+06	3.724E+04	24.8			
i	240	120	9.8	6.2	294.7	3912	3.891E+07	3.243E+05	99.7	2.835E+06	4.725E+04	26.9			
j	270	135	10.2	6.6	346.2	4595	5.790E+07	4.289E+05	112.3	4.197E+06	6.218E+04	30.2			
k	300	150	10.7	7.1	405.4	5381	8.356E+07	5.571E+05	124.6	6.036E+06	8.048E+04	33.5			
l	330	160	11.5	7.5	471.7	6261	1.177E+08	7.131E+05	137.1	7.878E+06	9.848E+04	35.5			
m	360	170	12.7	8.0	547.9	7273	1.627E+08	9.036E+05	149.5	1.043E+07	1.227E+05	37.9			
n	400	180	13.5	8.8	642.0	8521	2.321E+08	1.161E+06	165.1	1.317E+07	1.464E+05	39.3			
o	450	190	14.6	9.4	744.5	9882	3.374E+08	1.500E+06	184.8	1.675E+07	1.763E+05	41.2			
p	500	200	16.0	10.2	870.4	11552	4.820E+08	1.928E+06	204.3	2.141E+07	2.141E+05	43.1			
q	550	210	17.2	11.1	1013	13442	6.712E+08	2.441E+06	223.5	2.667E+07	2.540E+05	44.5			
r	600	220	19.0	12.0	1175	15598	9.208E+08	3.069E+06	243.0	3.386E+07	3.078E+05	46.6			
Notes: ¹ <i>I</i> = moment of inertia; <i>S</i> = section modulus; <i>r</i> = radius of gyration, used for column analysis; See sketch for X-X and Y-Y axes.															
² IPE sections have medium-width flanges. Narrow and wide-flange sections also available.															
³ Using density of steel = 7680 kg/m ³ .															
Source for data for additional sizes: Parker Steel Company—Metric Sized Metals, Toledo, Ohio.															

Tubing: Square and rectangular
Tables 15–14 to 15–16



U.S. Units

TABLE 15–14 Hollow Tubing: Steel, Square and Rectangular, Standard Structural HSS Shapes: 2 in to 8 in Depth

Design wall				1Section properties									
Shape		Thickness t_w (in)	2Weight per foot (lb/ft)	Area A (in ²)	Axis X-X			Axis Y-Y			Torsional constants		
Ref.	(in)				(in)	I_x (in ⁴)	S_x (in ³)	r_x (in)	I_y (in ⁴)	S_y (in ³)	r_y (in)	J (in ⁴)	C (in ³)
a	HSS 8×8× $\frac{1}{2}$		0.465	48.7	13.5	125	31.2	3.04	125	31.2	3.04	204	52.4
b	HSS 8×8× $\frac{1}{4}$		0.233	25.8	7.10	70.7	17.7	3.15	70.7	17.7	3.15	111	28.1
c	HSS 8×4× $\frac{1}{2}$		0.465	35.1	9.74	71.8	17.9	2.71	23.6	11.8	1.56	61.1	24.4
d	HSS 8×4× $\frac{1}{4}$		0.233	19.0	5.24	42.5	10.6	2.85	14.4	7.21	1.66	35.3	13.6
e	HSS 8×2× $\frac{1}{4}$		0.233	15.6	4.30	28.5	7.12	2.57	2.94	2.94	0.827	9.36	6.35
f	HSS 6×6× $\frac{1}{2}$		0.465	35.1	9.74	48.3	16.1	2.23	48.3	16.1	2.23	81.1	28.1
g	HSS 6×6× $\frac{1}{4}$		0.233	19.0	5.24	28.6	9.54	2.34	28.6	9.54	2.34	45.6	15.4
h	HSS 6×4× $\frac{1}{4}$		0.233	15.6	4.30	20.9	6.96	2.20	11.1	5.56	1.61	23.6	10.1
i	HSS 6×2× $\frac{1}{4}$		0.233	12.2	3.37	13.1	4.37	1.97	2.21	2.21	0.810	6.55	4.70
j	HSS 4×4× $\frac{1}{2}$		0.465	21.5	6.02	11.9	5.97	1.41	11.9	5.97	1.41	21.0	11.2
k	HSS 4×4× $\frac{1}{4}$		0.233	12.2	3.37	7.80	3.90	1.52	7.80	3.90	1.52	12.8	6.56
l	HSS 4×2× $\frac{1}{4}$		0.233	8.78	2.44	4.49	2.25	1.36	1.48	1.48	0.779	3.82	3.05
m	HSS 3×3× $\frac{1}{4}$		0.233	8.78	2.44	3.02	2.01	1.11	3.02	2.01	1.11	5.08	3.52
n	HSS 3×2× $\frac{1}{4}$		0.233	7.08	1.97	2.13	1.42	1.04	1.11	1.11	0.751	2.52	2.23
o	HSS 2×2× $\frac{1}{4}$		0.233	5.38	1.51	0.747	0.747	0.704	0.747	0.747	0.704	1.31	1.41

Notes: Example size: 6×4×1/4; 6 = vertical depth (in); 4 = width (in); 1/4 = wall thickness (in).

¹ I = moment of inertia; S = section modulus; r = radius of gyration.

²Using Density of steel = 0.283 lb_m/in³.

Sources for data for additional sizes: Metals Depot, Winchester KY.

Bull Moose Tube Co.

Steel Tube Institute of North America, Glenview, IL.